

GE Healthcare

1.5T Head Neck Spine (HNS)

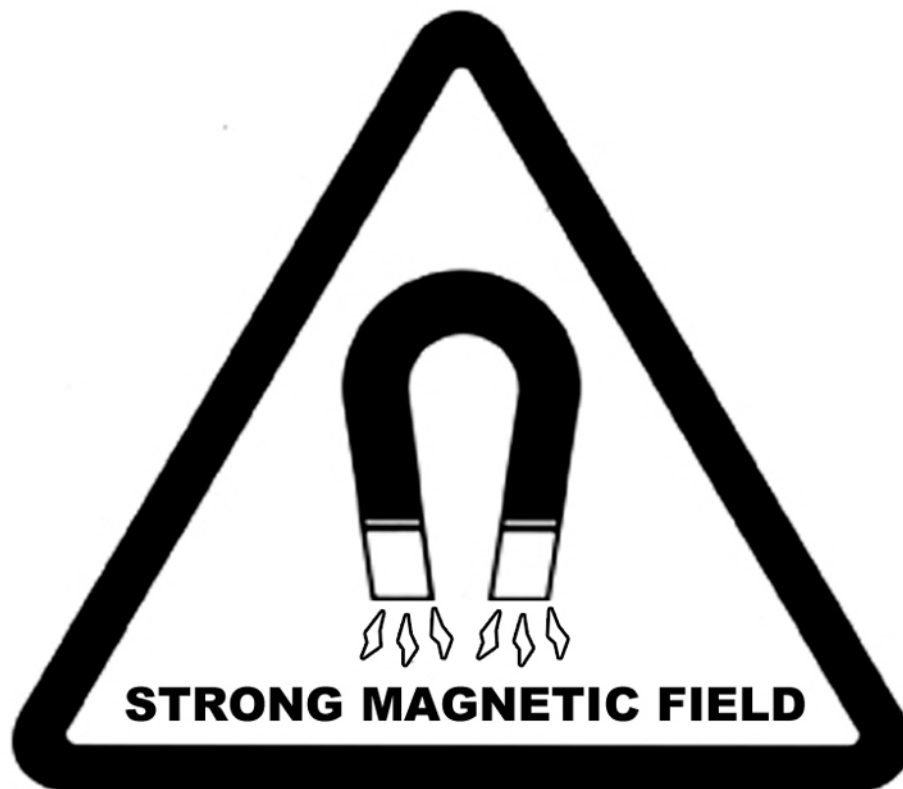
Service Methods



OPERATING DOCUMENTATION

2423983TPH
Revision 2

WARNING



NO PACEMAKERS

NO METALLIC IMPLANTS

Persons with pacemakers, neurostimulators or metallic implants must not enter the magnet area. Serious injury may result.



NO LOOSE METAL OBJECTS

Iron and steel materials must not be taken into the magnet area. Serious injury or property damage may result.

Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ

(BG)

Това упътване за работа е налично само на английски език с изключението на случаите, когато се изисква изрично по друг начин от местното законодателство, или от споразумение на местно ниво.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告

(ZH-CN)

本维修手册仅提供英文版本，除非当地法律或本地供应协议另有明确要求。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

Tento provozní návod existuje pouze v anglickém jazyce, pokud není jinak výslovně vyžadováno místními zákony nebo odsouhlaseno na místní úrovni.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

Denne servicemanual findes kun på engelsk, medmindre andet kræves i henhold til lokal lovgivning eller lokal aftale.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar, tenzij expliciet vereist door plaatselijke regelgeving of overeengekomen op lokaal niveau.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING

(EN)

This service manual is available in english only except as otherwise expressly required by local law or agreed to at a local level.

- If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

HOIATUS

(ET)

See teenindusjuhend on saadaval ainult inglise keeles, kui kohalikud seadused ei ütle teisiti või kui kohalikes õigusaktides ei ole otseselt teisiti ette nähtud.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

Tämä huolto-ohje on saatavilla vain englanniksi, ellei paikallinen laki nimenomaan toisin vaadi tai jos toisin on sovittu paikallisella tasolla.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

Sauf exigence contraire des lois locales ou accord contraire au niveau local, ce manuel d'installation et de maintenance n'est disponible qu'en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

Diese Serviceanleitung existiert nur in englischer Sprache, sofern nichts anderes gesetzlich vorgeschrieben oder auf lokaler Ebene vereinbart wurde.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά, εκτός αν η τοπική νομοθεσία απαιτεί κάτι άλλο ή υπάρχει διαφορετική συμφωνία σε τοπικό επίπεδο.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el, kivéve ha a helyi rendelkezések ezt kifejezetten elő nem írják, illetve ha helyi szinten erről külön megállapodás nem születik.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

Þessi þjónustuhandbók er eingöngu fánleg á ensku, nema annað sé sérstaklega krafist, löglega eða samþykkt á landsgrundvelli.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

Il presente manuale di manutenzione è disponibile soltanto in inglese, eccetto quando espressamente richiesto dalle normative locali o convenuto a livello locale.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告

(JA)

このサービスマニュアルには英語版しかありません。ただし使用国の法令に別異の定めがある、あるいは現地で別段の合意がある場合を除きます。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

현지 법률에 따라 명시적으로 요구하거나 현지 수준에서 합의한 경우를 제외하고 본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS

(LV)

Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā, izņemot gadījumus, kad vietējie likumi nepārprotami nosaka citādi vai panākta vienošanās vietējā līmenī.

- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

Šis eksploatavimo vadovas yra tik anglų kalba, išskyrus tuos atvejus, kai vietiniai įstatymai tiesiogiai numato kitokius reikalavimus arba vietinių lygiu sutarta kitaip.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL

(NO)

Denne servicehåndboken finnes bare på engelsk, bortsett fra dersom det motsatte uttrykkelig er fastsatt av lokal lovgivning eller det er inngått annen avtale lokalt.

- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim, chyba że lokalne przepisy lub umowy wyraźnie stanowią inaczej.

- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

ATENȚIE

(RO)

Acest manual de service este disponibil numai în limba engleză, cu excepția cazului în care este o cerință obligatorie stipulată de legislația națională sau convenită la nivel local.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

Данное руководство по техническому обслуживанию предлагается только на английском языке, за исключением тех случаев, когда наличие руководства на национальном языке является требованием местного законодательства или когда выпуск такого руководства согласован с местным представительством.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

UPOZORNENIE

(SK)

Tento návod na obsluhu je k dispozícii len v angličtine, okrem prípadov, kedy tak výslovne vyžadujú miestne zákony alebo je dohodnuté na miestnej úrovni.

- Ak zákaznikov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

ATENCION

(ES)

Este manual de servicio sólo existe en inglés, salvo que la legislación local exija de forma expresa lo contrario, o así se haya acordado a nivel local.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

Den här servicehandboken finns bara tillgänglig på engelska om inte annat uttryckligen krävs av lokal lag eller har överenskommit på lokal nivå.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DİKKAT

(TR)

Aksi, yerel bir yasa tarafından açıkça gerekli görülmediği veya yerel bir seviyede kabul edilmediği takdirde, bu servis kılavuzunun sadece İngilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu İngilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

AVISO

(PT-BR)

Este manual de assistência técnica encontra-se disponível unicamente em inglês, salvo disposições em contrário previstas pela legislação local ou acordadas no âmbito local.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO

(PT-PT)

Este manual de assistência técnica só se encontra disponível em inglês, salvo requisição expressa pela legislação local ou acordo efectuado a nível local.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

UPOZORENJE

(SR)

Ovo servisno uputstvo je dostupno samo na engleskom jeziku, sem ako lokalni zakon to izričito zahteva ili je dogovoreno na lokalnom nivou.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

Revision History

Rev	Date	Description
1.0	11/7/08	Per Michael Kamm, updated the following HNS procedures: • 1.5T Head Neck Spine (HNS) Coil Introduction • 1.5T Head Neck Spine (HNS) Coil Field Replacement Units • 1.5T Head Neck Spine (HNS) Coil Cable Replacement • 1.5T Head Neck Spine (HNS) Coil Troubleshooting Per Michael Kamm, created new HNS procedures: • 1.5T Head Neck Spine (HNS) Coil Mirror Repair/Replacement • 1.5T and 3.0T HNS Coil MCQA Setup • Direction 2423983TPH HNS Coil Installation (standalone document)
2.0	4/23/09	Updated the following: • Updated Illustration 1-1 • Updated contents for HNS Coil, Rev 2 and corrected part numbers for HNU posterior, HNU anterior, and NCU • Added HNS Rev 3 Discovery MR system serial labels • Updated HNS Coil Rev 3, adding Discovery MR • Updated Troubleshooting to include P-connector cable FRU part number, instructions on connecting the P-connector cable to the LPCA, and added the output cable RF/DC check for Discovery MR P-connector cable • Updated "Replacement" section to include Discovery FRU cable information and retitled the heading • Added a chapter on HNS Coil Setup for Reversed Ramped System

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Chapter 1 Introduction

1 1.5T Head Neck Spine (HNS) Coil Introduction

1.1 Product Identification and Shipping List

HNS Rev 02 Coil:

To identify the 1.5T Head, Neck and Spine Coil, refer to the coil labels (see [Illustration 1-1](#)).

NOTE: Each of the coils (5) subcomponents contains it own label.

Illustration 1-1: Coil Labels

GEHC COILS		Manufactured in the U.S.A. by: USA Instruments, Aurora, Ohio 44202	
HD HEAD, NECK AND SPINE ARRAY		NECK-CHEST UNIT	FRU Rev. 2
COIL P/N 2416329		FRU P/N 2417445	
COIL S/N FRU		FRU S/N N301	
COIL Rev. 2	Field 1.5T	1.5T 16 CHANNEL BRAIN-SPINE (Neurovascular) ARRAY COIL	
Frequency (MHz) 63.86		Date of Manufacture	

GEHC COILS		Manufactured in the U.S.A. by: USA Instruments, Aurora, Ohio 44202	
HD HEAD, NECK AND SPINE ARRAY		HNU ANTERIOR UNIT	FRU Rev. 2
COIL P/N 2416329		FRU P/N 2417362	
COIL S/N FRU		FRU S/N A360	
COIL Rev. 2	Field 1.5T	1.5T 16 CHANNEL BRAIN-SPINE (Neurovascular) ARRAY COIL	
Frequency (MHz) 63.86		Date of Manufacture	

GEHC COILS		Manufactured in the U.S.A. by: USA Instruments, Aurora, Ohio 44202	
HD HEAD, NECK AND SPINE ARRAY		HNU POSTERIOR UNIT	FRU Rev. 2
COIL P/N 2416329		FRU P/N 2417422	
COIL S/N FRU		FRU S/N P360	
COIL Rev. 2	Field 1.5T	1.5T 16 CHANNEL BRAIN-SPINE (Neurovascular) ARRAY COIL	
Frequency (MHz) 63.86		Date of Manufacture	

GEHC COILS		Manufactured in the U.S.A. by: USA Instruments, Aurora, Ohio 44202	
HD HEAD, NECK AND SPINE ARRAY		ANT. ADAPTER UNIT	FRU Rev. 2
COIL P/N 2416329		FRU P/N 2417441	
COIL S/N FRU		FRU S/N B360	
COIL Rev. 2	Field 1.5T	1.5T 16 CHANNEL BRAIN-SPINE (Neurovascular) ARRAY COIL	
Frequency (MHz) 63.86		Date of Manufacture	

GEHC COILS		Manufactured in the U.S.A. by: USA Instruments, Aurora, Ohio 44202	
HD HEAD, NECK AND SPINE ARRAY		T/L SPINE UNIT	FRU Rev. 1
COIL P/N 2416329		FRU P/N 2417361	
COIL S/N FRU		FRU S/N S360	
COIL Rev. 2	Field 1.5T	1.5T 16 CHANNEL BRAIN-SPINE (Neurovascular) ARRAY COIL	
Frequency (MHz) 63.86		Date of Manufacture	

Illustration 1-2: The HDx 1.5T Head, Neck And Spine Coil



SHIPPING CONTENTS / PACKAGING LIST : 3 CONNECTOR COIL FOR UPGRADE SYSTEM

Table 1-1: Box 1 Content

Part Name	Part Number	Qty.
Head unit - posterior	2417438	1
Head unit – anterior or face piece	2417422	1
TL spine unit	2417444	1
Chest piece	2417445	1
Horse shoe	2417441	1
Patient comfort pad - transition	2417243	1
Right shoulder pad	2417244	1
Left shoulder pad	2417245	1
Patient comfort pad - TL	2417246	1
Patient comfort pad - Head	2417247	1
Wedge pad	2417649	2
Head unit cable – B conn (attached to coil)	2417443	1
Head unit cable – legacy (attached to coil)	2417446	1
TL cable (attached to coil)	2417442	1
Operator manual	2417746	1

Table 1-2: Box 2 Content

Part Name	Part Number	Qty.
Phantom kit – head section and positioner	2418236 (2420112 - Positioner, 2424360 - Head Phantom, 2422888 - Neck Phantom)	1

Table 1-3: Box 3 Content

Part Name	Part Number	Qty.
Phantom kit – TL component	2424359 (2 each of TL phantom: 2424361)	1

SHIPPING CONTENTS / PACKAGING LIST : 2 CONNECTOR COIL FOR FORWARD PRODUCTION SYSTEM**Table 1-4: Box 1 Content**

Part Name	Part Number	Qty.
Head unit - posterior	2417438	1
Head unit – anterior or face piece	2417422	1
TL spine unit	2417444	1
Chest piece	2417445	1
Horse shoe	2417441	1
Patient comfort pad - transition	2417243	1
Right shoulder pad	2417244	1
Left shoulder pad	2417245	1
Patient comfort pad - TL	2417246	1
Patient comfort pad - Head	2417247	1
Wedge pad	2417649	2
Head unit cable labeled B (attached to coil)	2417440	1
TL cable (attached to coil)	2417442	1
Operator manual	2417746	1

Table 1-5: Box 2 Content

Part Name	Part Number	Qty.
Phantom kit – head section and positioner	2418236 (2420112 - Positioner, 2424360 - Head Phantom, 2422888 - Neck Phantom)	1

Table 1-6: Box 3 Content

Part Name	Part Number	Qty.
Phantom kit – TL component	424359 (2 each of TL phantom: 2424361)	1

Illustration 1-3: THE HDx AND DISCOVERY SYSTEMS 1.5T HEAD, NECK AND SPINE COIL



SHIPPING CONTENTS / PACKAGING LIST : 3 CONNECTOR COIL FOR UPGRADE SYSTEM

Table 1-7: Box 1 Content

Part Name	Part Number	Qty.
HNU Posterior Unit	2423192	1
HNU Anterior Unit		
ANT. Adapter Unit		
TL spine unit	2423194	1
Neck-Chest Unit	2423193	1
Patient comfort pad - transition	2417243	1
Right shoulder pad	2417244	1
Left shoulder pad	2417245	1
Patient comfort pad - TL	2417246	1
Patient comfort pad - Head	2417247	1
Wedge pad	2417649	2
Head unit cable – B conn (attached to coil)	2423004	1
Head unit cable – legacy (attached to coil)	2423006	1
TL cable (attached to coil)	2423005	1
Operator manual	2423588-500	1

Table 1-8: Box 2 Content

Part Name	Part Number	Qty.
Phantom kit – head section and positioner	2418236 (2420112 - Positioner, 2424360 - Head Phantom, 2422888 - Neck Phantom)	1

Table 1-9: Box 3 Content

Part Name	Part Number	Qty.
Phantom kit – TL component	2424359 (2 each of TL phantom: 2424361)	1

SHIPPING CONTENTS / PACKAGING LIST : 2 CONNECTOR COIL FOR FORWARD PRODUCTION SYSTEM

Table 1-10: Box 1 Content

Part Name	Part Number	Qty.
HNU Posterior Unit	2423192	1
HNU Anterior Unit		
ANT. Adapter Unit		
TL spine unit	2423194	1
Neck-Chest Unit	2423193	1
Patient comfort pad - transition	2417243	1
Right shoulder pad	2417244	1
Left shoulder pad	2417245	1
Patient comfort pad - TL	2417246	1
Patient comfort pad - Head	2417247	1
Wedge pad	2417649	2
Head unit cable labeled B (attached to coil)	2423003	1
TL cable (attached to coil)	2423005	1
Operator manual	2423588-500	1

Table 1-11: Box 2 Content

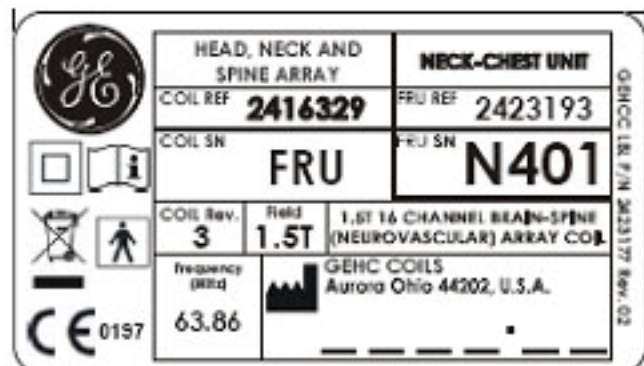
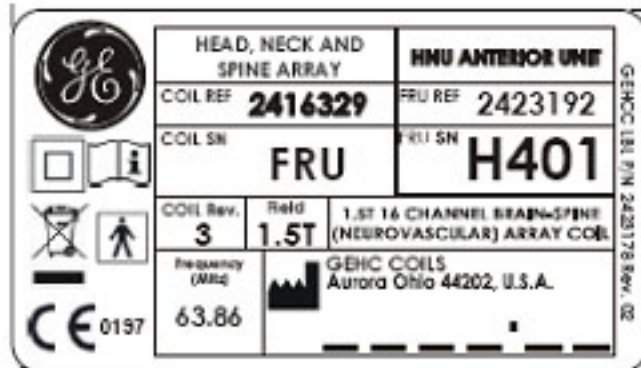
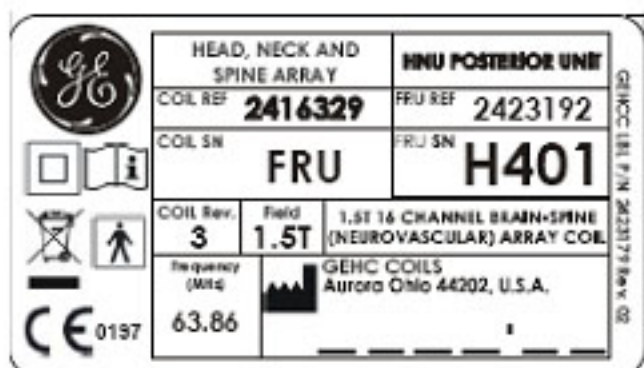
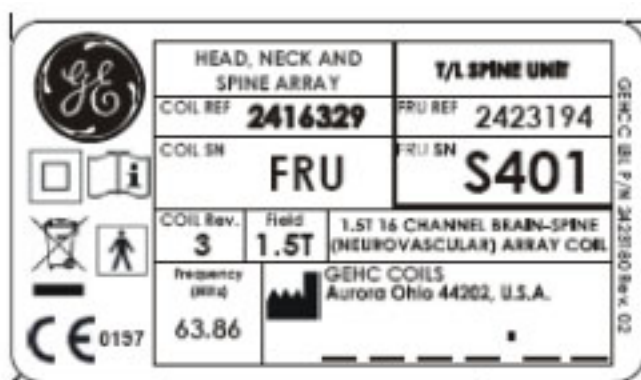
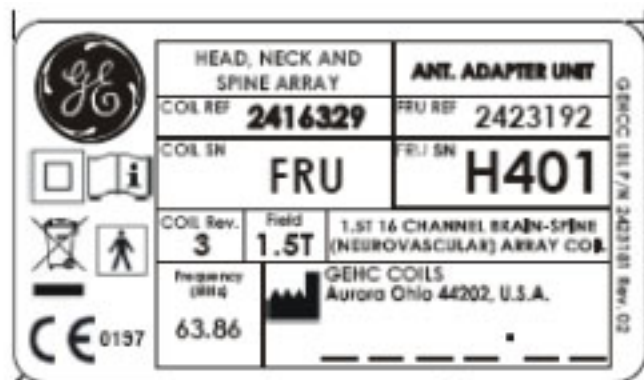
Part Name	Part Number	Qty.
Phantom kit – head section and positioner	2418236 (2420112 - Positioner, 2424360 - Head Phantom, 2422888 - Neck Phantom)	1

Table 1-12: Box 3 Content

Part Name	Part Number	Qty.
Phantom kit – TL component	2424359 (2 each of TL phantom 2424361)	1

HNS REV 3 FOR DISCOVERY 450 SYSTEM

Illustration 1-4: HNS Rev 3 Coil Labels



SHIPPING CONTENTS / PACKAGING LIST : 2 CONNECTOR COIL DISCOVERY SYSTEM

Table 1-13: Box 1 Content

Part Name	Part Number	Qty.
HNU Posterior Unit	2423192	1
HNU Anterior Unit		
Anterior Adapter Unit		
TL spine unit	2423194	1
Neck-Chest Unit	2423193	1
Patient comfort pad - transition	2417243	1
Right shoulder pad	2417244	1
Left shoulder pad	2417245	1
Patient comfort pad - TL	2417246	1
Patient comfort pad - Head	2417247	1
Wedge pad	2417649	2
Head unit cable – P connector (attached to coil)	2423972	1
TL cable (attached to coil)	2423005	1
Operator manual	2423588-500	1

Table 1-14: Box 2 Content

Part Name	Part Number	Qty.
Phantom kit – head section and positioner	2418236 (2420112 - Positioner, 2424360 - Head Phantom, 2422888 - Neck Phantom)	1

Table 1-15: Box 3 Content

Part Name	Part Number	Qty.
Phantom kit – TL Component	2424359 (2 each of TL phantom: 2424361)	1

1.2 Compatibility

The Head, Neck and Spine coil is compatible with the HDx and Discovery 450 1.5T system.

1.3 Related Documentation

Operator's Guide, GE Part Number 2417746 for HNS Rev 02 coil and 2423588-500 for HNS Rev 03 coil.

1.4 Environmental Requirements

Storage Requirements

Coil should be stored in the scanner room.

1.5 Accessories

Table 1-16: Accessories

Part Number	Description
2418076	TL phantom kit
2418078	HN phantom Kit
2418081	Pad kit

Chapter 2 Adjustment and Calibration

1 1.5T Head Neck Spine (HNS) Coil MCQA Setup

1.1 Personnel Requirements

Required Persons	Procedure
1	30

1.2 Overview

Follow this process to prepare for the MCQA test.

NOTE: The coil consists of five units: Posterior Head, Face, Chest, Horseshoe, and TL. This is done to ensure modularity in design and diagnostics. The MCQA test for the coil is conducted to three configurations: TL unit only, Posterior Head Chest Horseshoe TL, and Posterior Head Face.

1.3 Preliminary Requirements

1.3.1 Tools and Test Equipment

No special tools or test equipment required.

1.3.2 Safety

No special safety requirement.

1.3.3 Required Conditions

The appropriate coil configuration file must be installed to run this tool.

1.4 Procedure

NOTE: If installing the coil for the first time in a system, please refer to Auto Coil Install.

NOTE: 1.5T phantoms are green.

1. To position the coil in the cradle, connect the cable as shown in [Illustration 2-1](#):

Illustration 2-1: Connecting the Coil



2. Remove the pads and place a phantom in the coil as shown in [Illustration 2-2](#):

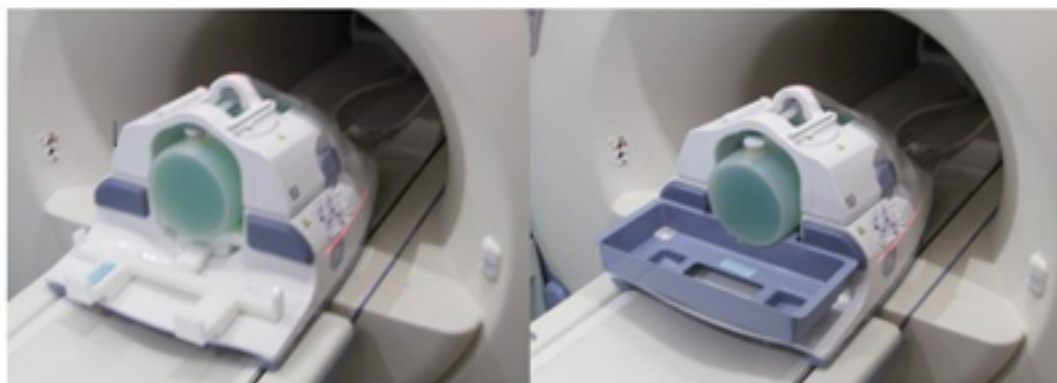
Illustration 2-2: Positioning Phantom for the TL unit, Posterior Head Chest Horseshoe TL and Posterior Head Face Configurations of the HNS Coil



Positioning the phantoms for TL Unit



Positioning the phantoms for Posterior Head Chest Horseshoe TL



Positioning the phantom for Posterior Head Face

NOTE: The MCQA tool can ONLY run on these individual coil units and NOT all coil units at the same time in combination. MCQA will not run and will report that the configuration is invalid if an attempt is made to scan with an unsupported combination of coils.

3. Landmark at the positions shown in [Illustration 2-3](#), [Illustration 2-4](#), and [Illustration 2-5](#), as the case may be.

Illustration 2-3: TL Only Setup Landmarking

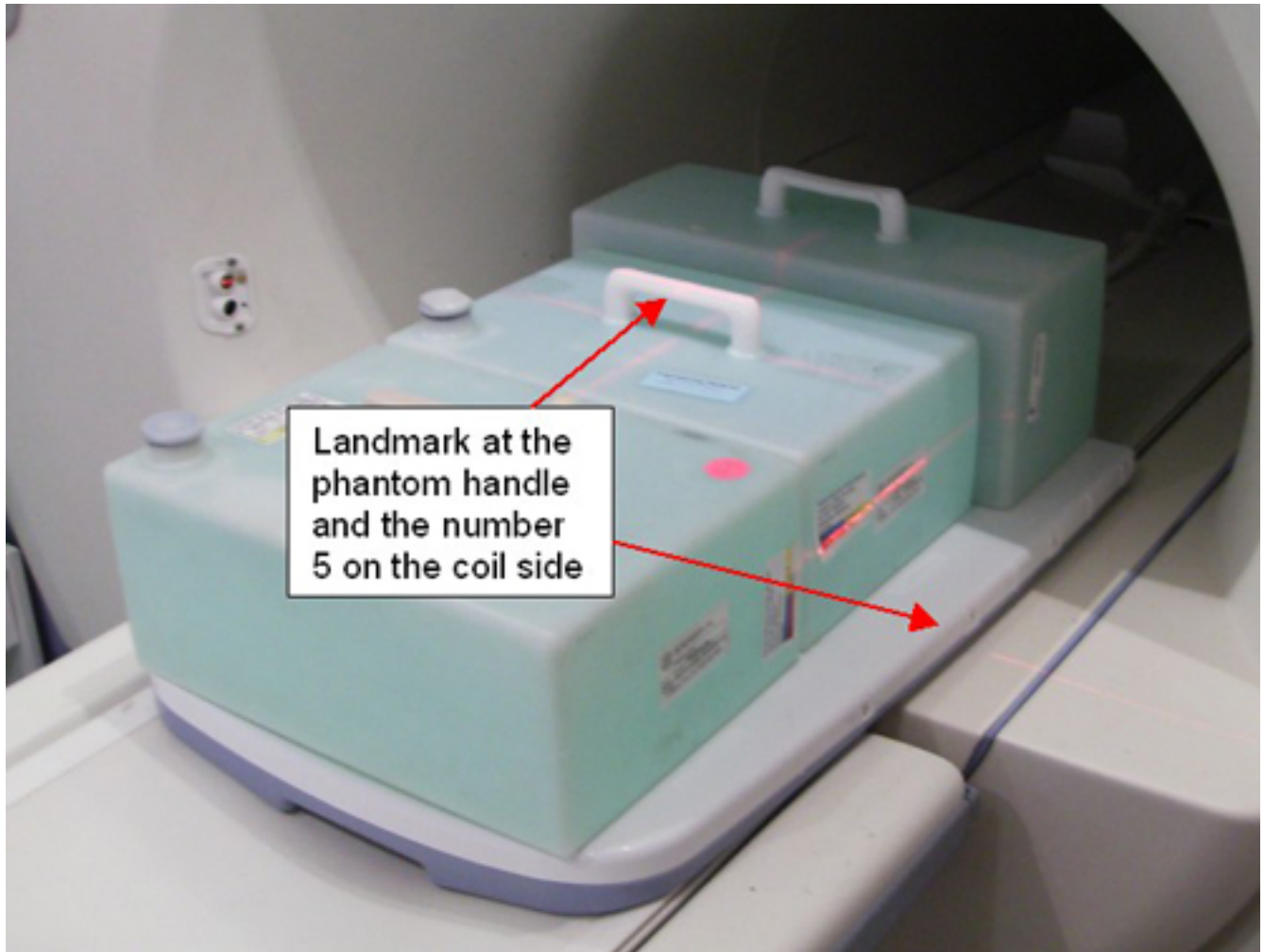


Illustration 2-4: Posterior Head Chest Horseshoe TL Setup Landmarking

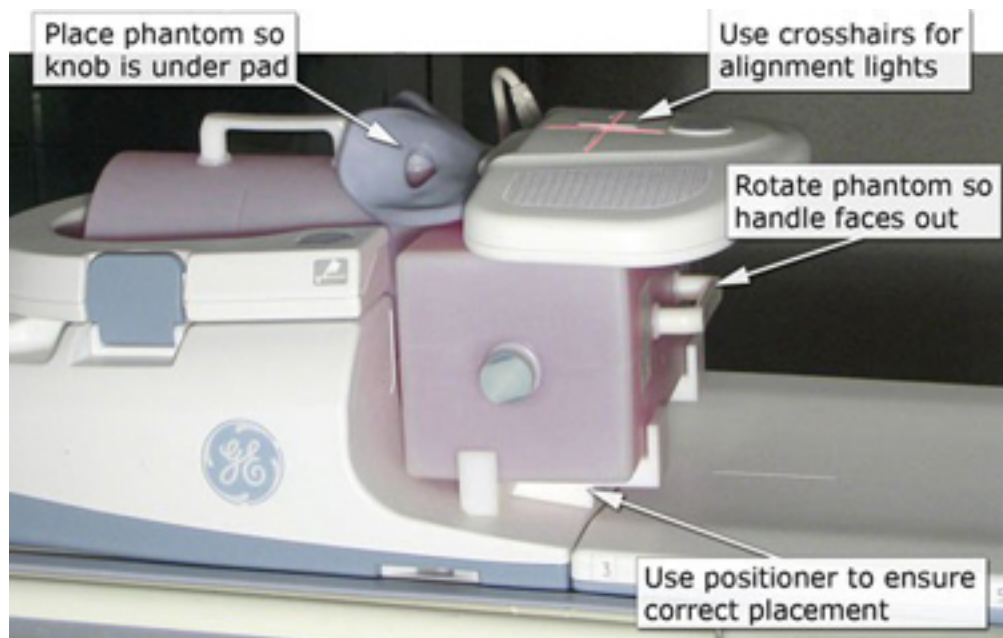
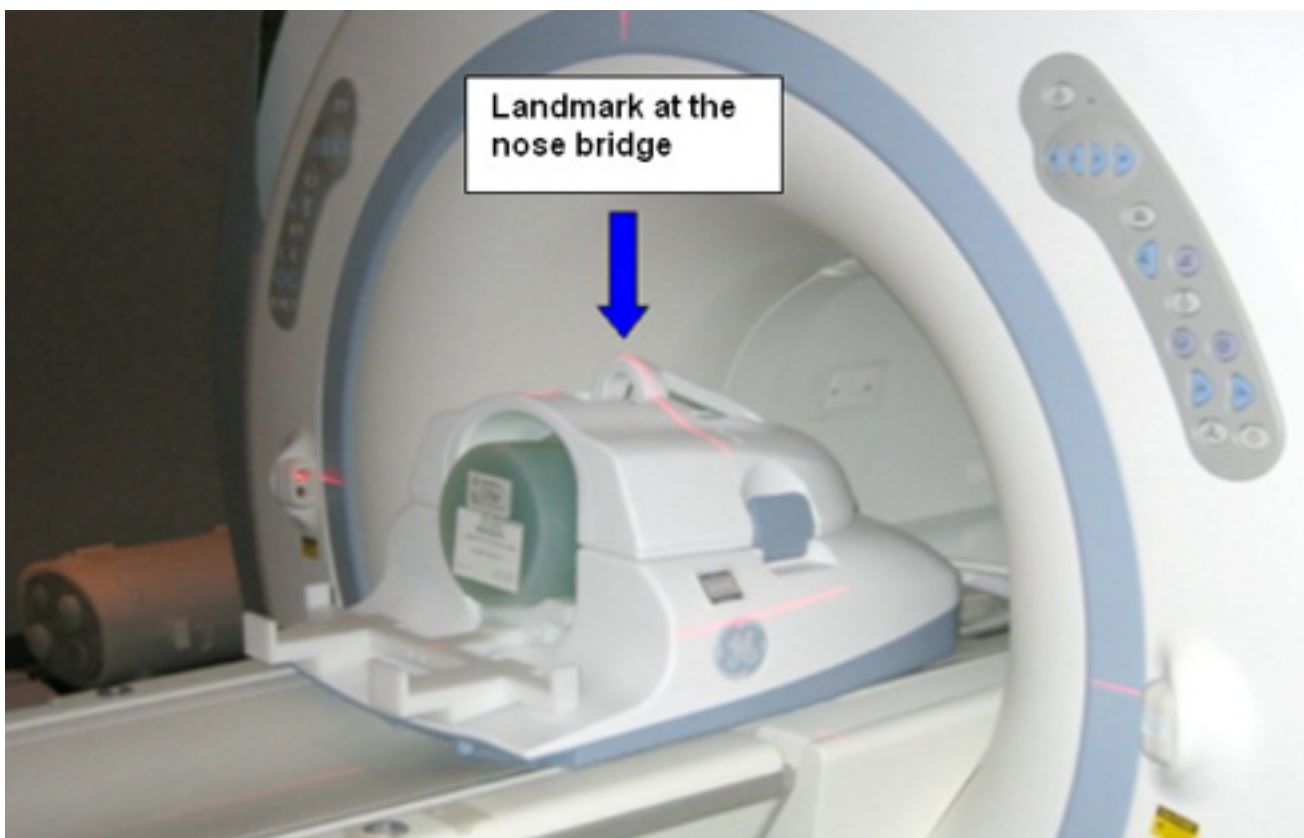


Illustration 2-5: Posterior Head Face Setup Landmarking



1.5 Perform the Surface Coil MCQA Test

See MCQA test instructions for the next steps required to perform the MCQA test.

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Chapter 3 Troubleshooting

1 1.5T Head Neck Spine (HNS) Coil Troubleshooting

1.1 Personnel Requirements

Required Persons	Procedure
1	30 min

1.2 Overview

The tips in this document can be used to troubleshoot common problems with the 1.5T HNS coil.

1.3 Preliminary Requirements

1.3.1 Tools and Test Equipment

Some tests may require Phantoms for IQ Testing.

Digital Multi Meter.

1.3.2 Replacement Parts

HNS Rev 2		
Part Number	FRUs	Accessories
2417362	Head Neck Unit (A and P)	
2418074	HNU Adapter Block	
2417360	Neck Chest Unit	
2417361	TL Unit	
2417356	Upgrade Output Cable-Hypertronics	
2417357	Upgrade Output Cable-bendix	
2417359	Forward Production Output Cable	
2417358	TL Unit Cable	
2418080	Mirror	
2418076	TL Phantom Kit	Yes
2418078	HN Phantom Kit	Yes
2418081	Pad Kit	Yes

HNS Rev 3		
Part Number	FRUs	Accessories
2423192	Head Neck Unit 1.5T Brain Spine FRU	No
2423193	Neck Chest Unit 1.5T BRAIN SPINE FRU	No
2423194	TL Unit 1.5T Brain Spine FRU	No
2423319	8 Ch HNS EXC III Upgrade System Cable FRU	No
2423317	8 Ch HNS EXC II System Cable FRU	No
2423320	16Ch EXC III HNS Forward Production System Cable FRU	No
2424269	16ch HNS key-connector system cable FRU	No
2423318	HNS TL Unit Cable FRU	No
2419521	Mirror Assembly, FRU, 1.5T	Yes
2418076	TL Phantom Kit	Yes
2418078	HN Phantom Kit	Yes
G3335RK	HNS Phantom Kit	Yes
2418081	Pad Kit	Yes

1.3.3 Required Conditions

The following coil configurations must be installed to run the MCQA test for the coil:

HNS Head
HNS BrachPlex
HNS CTLBOT

Refer to the operator manual for all coil configurations.

1.4 Troubleshooting Overview

Refer to the following sections when troubleshooting the HNS coil:

- Receiving No Signal - [Section 1.5.1](#)
- Image Quality - [Section 1.5.2](#)
- Artifacts - [Section 1.5.3](#)
- Output Cable Check - [Section 1.5.4](#)
 - Forward Production System HNS Rev 2 - [Section 1.5.4.1](#)
 - Forward Production System HNS Rev 3 - [Section 1.5.4.2](#)
 - Upgrade System HNS Rev 2 - [Section 1.5.4.3](#)
 - Upgrade System HNS Rev 3 - [Section 1.5.4.4](#)
 - Discovery MR450 System HNS Rev 3 - [Section 1.5.4.5](#)
 - Configuration of Jumpers on the Interface Board for Upgrade and Forward Production System - [Section 1.5.4.6](#)

1.5 Receiving No Signal

1.5.1 Receiving No Signal

Problem:

You are unable to pre-scan or are scanning, yet receiving no signal, and receiving part of signal from coil elements.

Possible Solution:

Verify that the port which is used to plug in the coil either has a green light illuminated (Port A or Port B) or a green lock (Port P1, P2, P3 or P4) indicated. This indicates the coil is properly plugged into the system and the coil I.D. on the coil is properly installed.

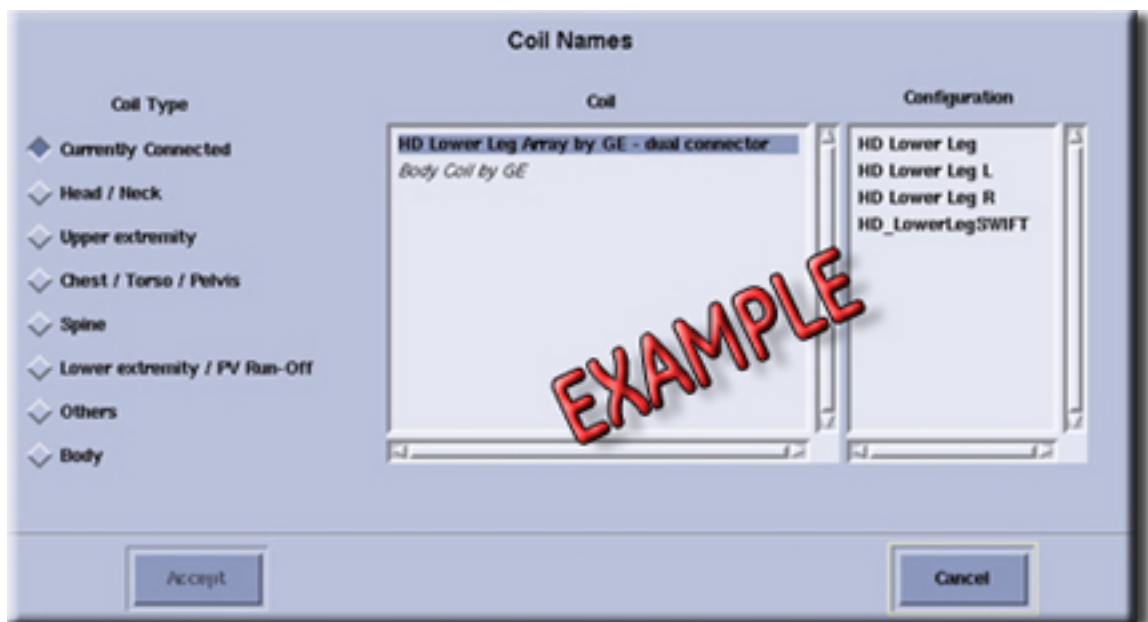
NOTE: For Forward Production systems, we use Port A (Hypertronics, 8 channels) for TL (Thoracic Lumbar), and Port B (Hypertronics, 16 Channel) for Head Neck. If the TL is used by itself, then it can be connected to port B too.

For upgrade systems: we use Port A (Hypertronics, 8 channels) for TL (Thoracic Lumbar), Port B (Hypertronics, 8 Channel) and 8 channel Bendix (Legacy) connector for Head Neck.

For Discovery systems, use Port P1 or Port P2 (P-connector, 16 channel) for HNU and use Port A (Hypertronics, 8 channel) for TL Unit.

Verify that the system has correctly detected the coil by checking in the “Currently Connected” in the GUI to select coils in Rx as shown in [Illustration 3-1](#).

Illustration 3-1: “Currently Connected” Coils Window In The Scan Screen



- Ensure the anterior or adaptor block is fully latched.

- If the Neck Chest unit is used, ensure its connector is fully engaged to the receptacle on the posterior Head Neck unit.
- Verify that the landmark is correct and that the cradle has not unlatched.
- Verify that the scan locations and any FOV offsets are correct.
- Perform a continuity check on the output cable ([Section 1.5.4](#)) if you lose some signals from some of coil channels. The continuity check must be performed by a GE authorized Service Engineer.
- Verify that the coil is positioned with the cable exiting towards the bore.

If you still cannot get a signal, try to scan (transmit and receive) with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still receive no signal the problem probably lies with the MR system. If the scan completes successfully, there is probably a problem with the Brain Spine coil. Contact GE for further assistance. If you are unable to scan with the substitute coil, there may be a system problem related to this particular coil type.

1.5.2 Image Quality

Problem:

Poor IQ, shaded images, or if MCQA fails

Possible Solution:

- Perform a continuity check on the output cable ([Section 1.5.4](#)). The continuity check must be performed by a GE authorized Service Engineer.
- Verify that there are no loops in the cables.
- Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).
- Verify that the coil is properly positioned.
- Verify that your center frequency is within the frequency adjustment range for your system.
- Verify that the R1, R2 and TG values from the pre-scan are within normally expected ranges.

If you have not done so already, perform a system Quality Assurance MCQA test. If the values you obtain do not fall within normal operating parameters, investigate this further by performing a phantom scan with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still have the same problems, there is probably an MR system problem. If the body coil scan is satisfactory, acquire a scan using both another coil of the same type (receive-only, phased array or transmit/receive, whichever applies) and the same system coil selection. If the image quality is visibly improved, there may be a problem with the Head Neck Spine coil. Contact GE for further assistance. If the image quality still suffers, there may be a system problem related to imaging with this type of coil.

1.5.3 Artifacts

Problem:

There is a black line, strip lines, or signal void on the image.

Possible Solution:

- Verify that there is no metal present in the area being scanned in or on the patient.
- Verify that no hair is caught trapped in the connectors between the anterior and posterior sections of the Head Neck unit. This may appear as an intermittent oscillations.
- Verify the cable connector of the Neck Chest unit is fully plugged into the receptacle on the posterior section of the Head Neck unit. If the connector is not fully plugged in, there may be a coil not present error or there may be intermittent oscillations.
- If you have oscillation of the preamp you will see striped lines.

1.5.4 Output Cable Check

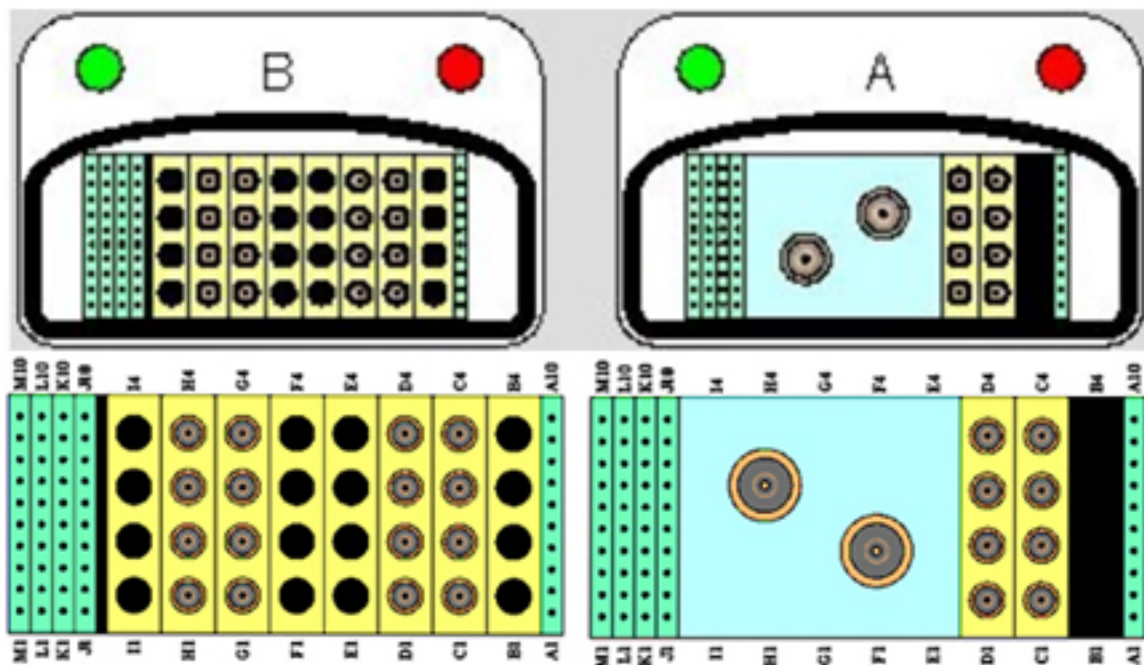
Before removing the cable assembly from the coil, visually inspect all coaxial connectors and pins on the system plug. If there are any broken, deformed or recessed pins or coaxial connectors, replace the cable assembly. When you replace the cable assembly, you need to make sure the configuration of jumpers on the interface board for upgrade and forward production systems.

Remove the cable assembly from the coil. See [Chapter 5, 1.5T HNS Coil Cable Replacement](#).

1.5.4.1 Forward Production System HNS Rev 2

We use 8Ch “A” Hypertronics connector cable for TL unit and 16Ch “B” Hypertronics connector cable for HN unit on forward production system.

Illustration 3-2: Pin Layouts Of Hypertronics Connectors (A and B) For Forward Production System



RF Signal Channels Check

NOTE: The center pins of the System Connector receive coaxial connectors are extremely fragile. Do not attempt to make any measurements at these center pins.

The following procedure will rule out a short to GND for each of the signal pins detailed in the following table and ensure the integrity of the ground connection for each RF coax cable. This check does not determine continuity for each of the RF signal connections.

With the coil cable disconnected, use a multimeter to assure no open circuit condition exists between the RF ground shields of the RF coax connector in the system side and the coils side (Refer to [Table 3-1](#) or [Table 3-2](#).) Use the multimeter to assure an open circuit condition exists between the center pin and the shield for each RF connector in the coil side.

Table 3-1: 8Ch “A” connector RF Allocation

System Side Connector RF GND Shield	Coil Side Connector RF Center Pin
C1	J4
C2	J5
C3	J6
C4	J7
D1	J8
D2	J9
D3	J10
D4	J11

Illustration 3-3: 8Ch “A” Connector RF Cables From Coils Side

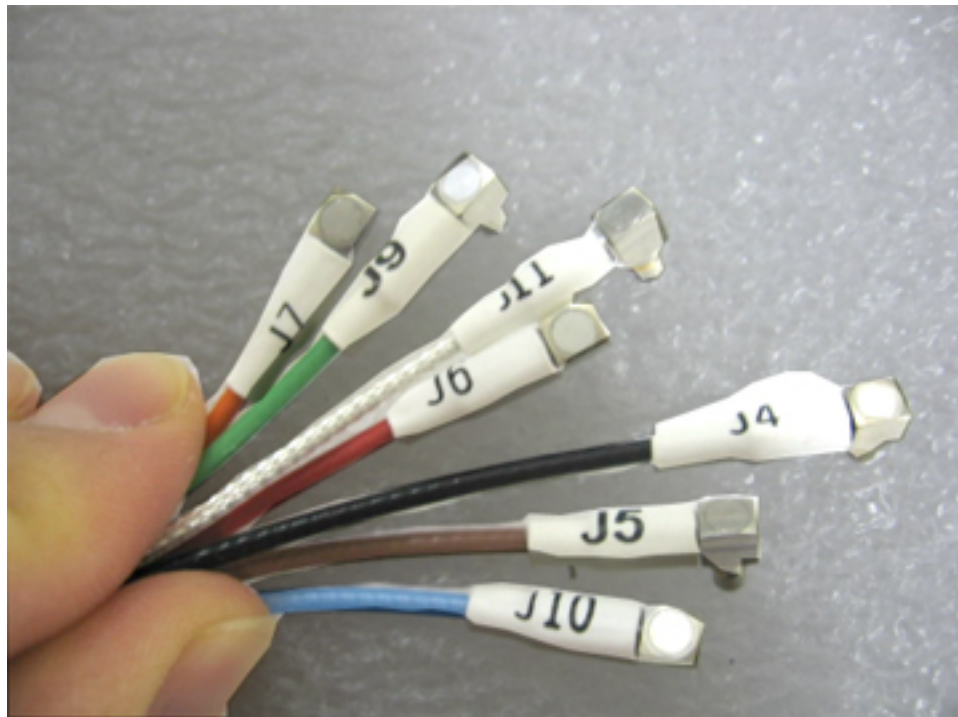
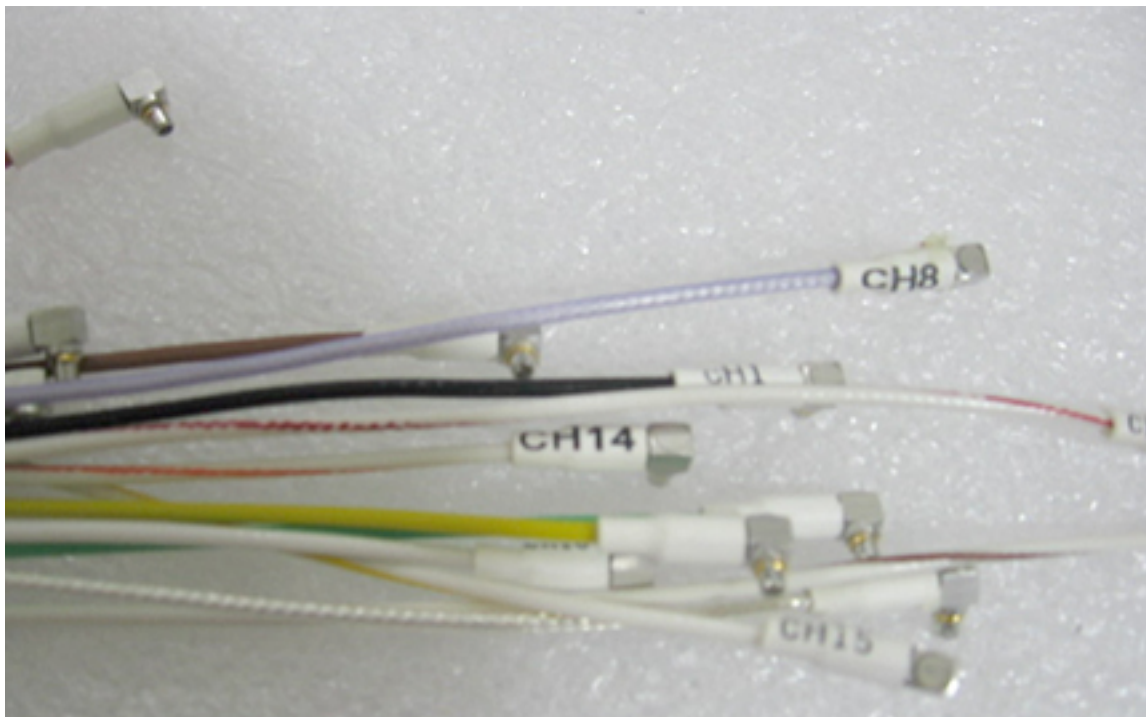


Table 3-2: 16Ch “B” connector RF Allocation

System Side Connector RF GND Shield	Coil Side Connector RF Center Pin
C1	CH1
C2	CH2
C3	CH3
C4	CH4
D1	CH5
D2	CH6
D3	CH7
D4	CH8
G1	CH9
G2	CH10
G3	CH11
G4	CH12
H1	CH13
H2	CH14
H3	CH15
H4	CH16

Illustration 3-4: 16Ch “B” Connector RF Cables From Coils Side



If one or more of the readings indicate a short, replace the cable. If all of the above readings indicate an open circuit condition, proceed to the DC Lines Continuity Check.

DC Lines Continuity Check

With the coil cable disconnected, use a multimeter to determine continuity between the designated pins for each control line below. Also test the continuity between the pairs of pins within system connector that are listed in the tables below. If any line or pair of pins indicates an open, replace the output cable.

Table 3-3: "A" connector DC Allocation

System Side Connector DC Pin	Coil Side Connector DC Center Pin
M3	J1-1
M6	J1-2
J1	J2-1
K1	J2-2
J2	J2-3
K2	J2-4
J3	J2-5
K3	J2-6
J4	J2-7
K4	J2-8
M7	J3-1
M8	J3-2
Short A1 to A2	N/A
Short J9 to J10	N/A
Short M9 to M10	N/A

Illustration 3-5: "A" Connector DC Cables From Coils Side

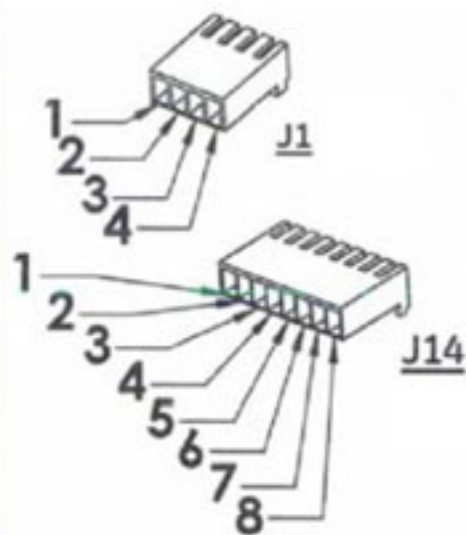
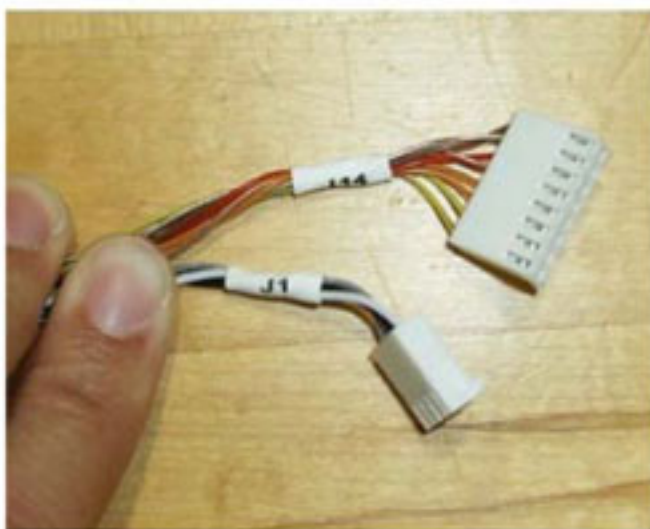
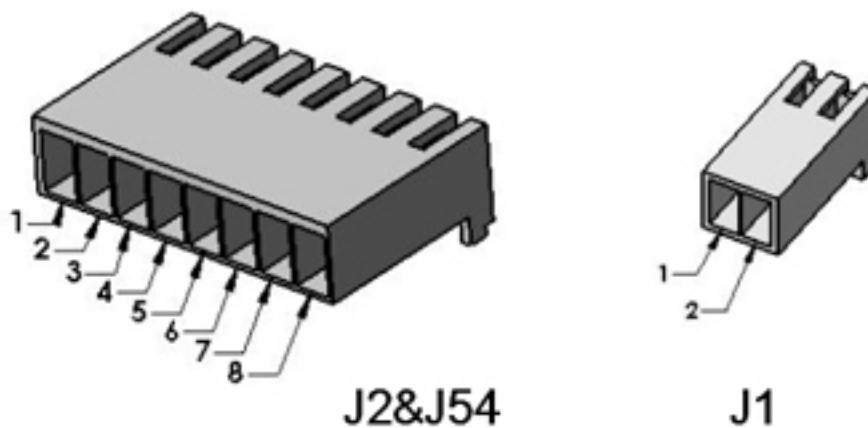
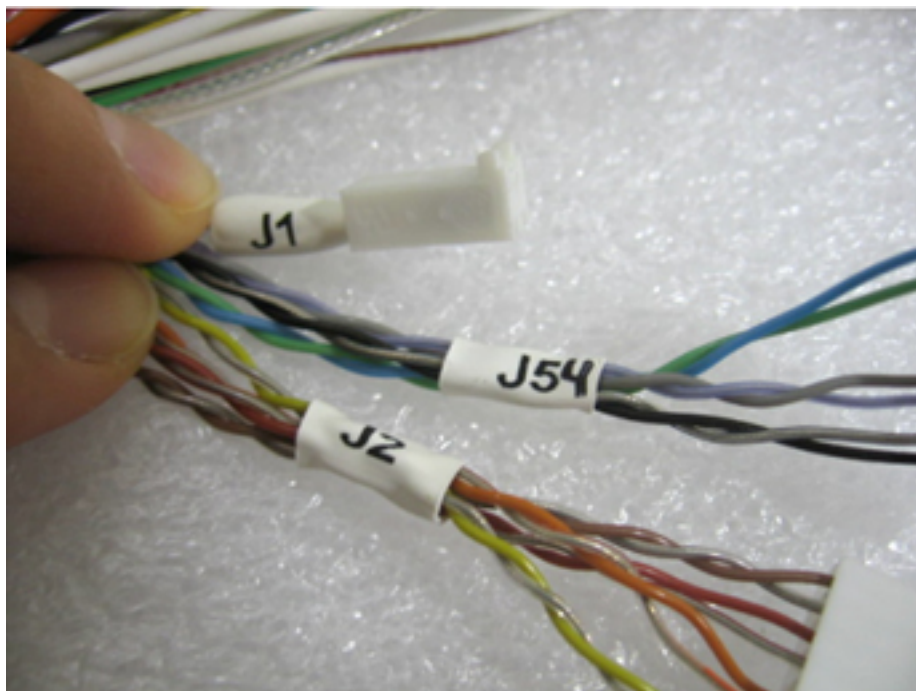


Table 3-4: "B" Connector DC Allocation

System Side Connector DC Pin	Coil Side Connector DC Pin
A1	J54-1
A2	J54-2
A3	J2-1
A4	J2-2
A5	J2-3
A6	J2-4
A7	J2-5
A8	J2-6
A9	J2-7
A10	J2-8
M2	J54-7
M5	J54-8
M6	J1-2
M7	J54-3
M8	J54-4
Short J9 to J10	N/A
Short M9 to M10	N/A

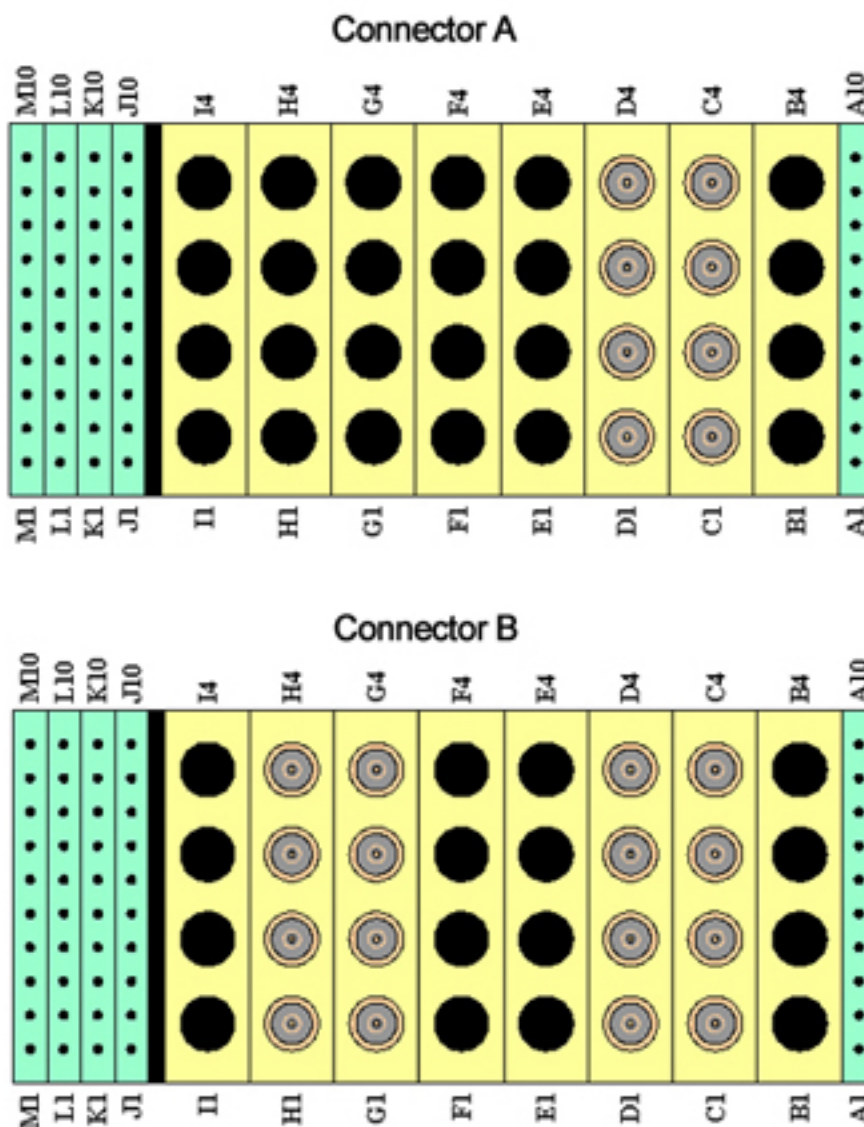
Illustration 3-6: "B" connector DC Cables From Coils Side



1.5.4.2 Forward Production System HNS Rev 3

We use 8Ch "A" Hypertronics connector cable for TL unit and 16Ch "B" Hypertronics connector cable for HN unit on forward production system.

Illustration 3-7: Pin Layouts Of Hypertronics Connectors (A and B) For Forward Production System

**RF Signal Channels Check**

NOTE: The center pins of the System Connector receive coaxial connectors are extremely fragile. Do not attempt to make any measurements at these center pins.

The following procedure will ensure the integrity of the ground connection for each RF coax cable detailed in the following tables.

With the coil cable disconnected, use a multimeter to assure an open circuit condition exists between the RF ground shields of the RF coax connector in the system side and the coils side.

Table 3-5: 8Ch "A" connector RF Allocation

System Side Connector RF GND Shield	Coil Side Connector RF GND Shield Pin # (All located on Gang connector J28)
C1	8
C2	1
C3	7
C4	2
D1	6
D2	3
D3	5
D4	4

Illustration 3-8: 8Ch "A" Connector RF Cables From Coils Side

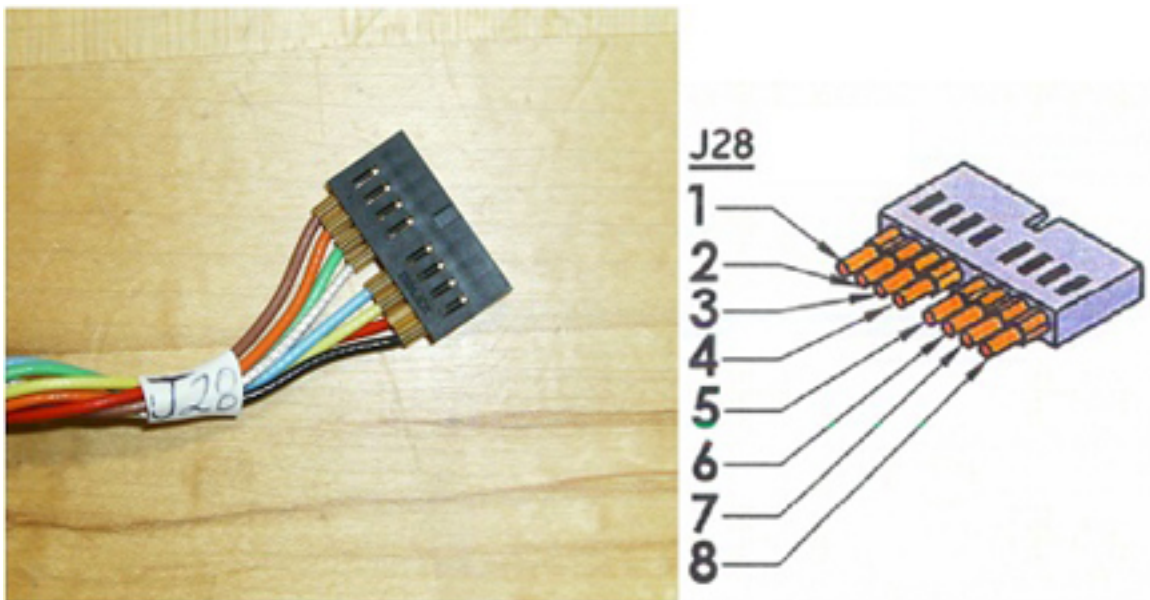
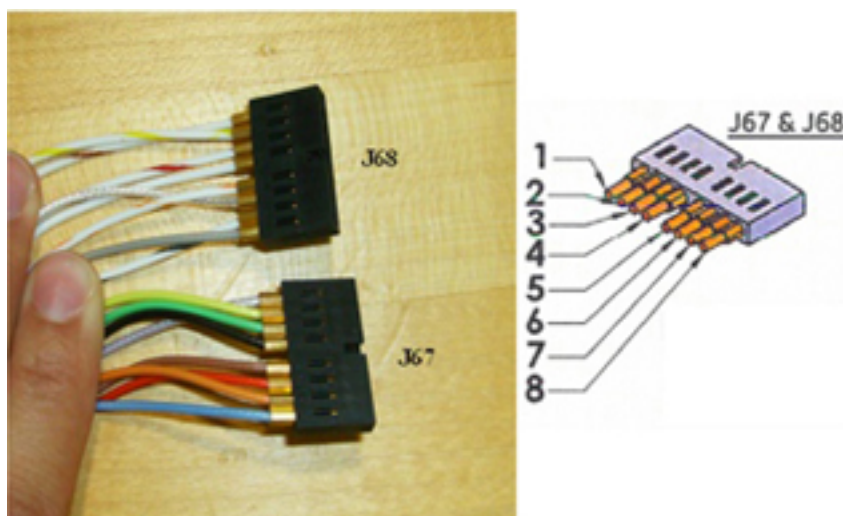


Table 3-6: 16Ch “B” connector RF Allocation

System Side Connector RF GND Shield	Coil Side Connector RF GND Shield Pin # (All located on Gang connector J67)	System Side Connector RF GND Shield	Coil Side Connector RF GND Shield Pin # (All located on connector J68)
C1	4	G1	8
C2	5	G2	4
C3	6	G3	7
C4	7	G4	1
D1	2	H1	3
D2	3	H2	5
D3	8	H3	2
D4	1	H4	6

Illustration 3-9: 16Ch “B” Connector RF Cables From Coils Side



If one or more of the readings indicate an open, replace the cable. If no open is found for the ground of the RF cables, proceed to the DC Lines Continuity Check.

DC Lines Continuity Check

With the coil cable disconnected, use a multimeter to determine continuity between the designated pins for each DC control line below. Also test the continuity between the pairs of pins within system connector that are listed in the tables below. If any line or pair of pins indicates an open, replace the output cable.

Table 3-7: "A" Connector DC Allocation

System Side Connector DC Pin	Coil Side Connector DC Pin
M3	J1-1
M6	J1-2
M8	J1-3
M7	J1-4
J1	J14-1
K1	J14-2
J2	J14-3
K2	J14-4
J3	J14-5
K3	J14-6
J4	J14-7
K4	J14-8
Short A1 to A2	
Short J9 to J10	N/A
Short M9 to M10	

Illustration 3-10: "A" Connector DC Cables From Coils Side

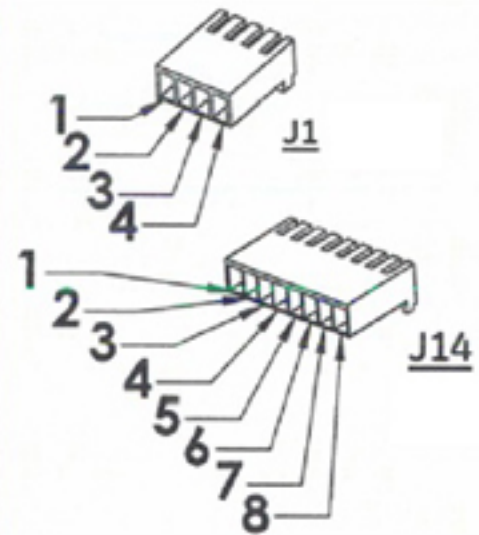
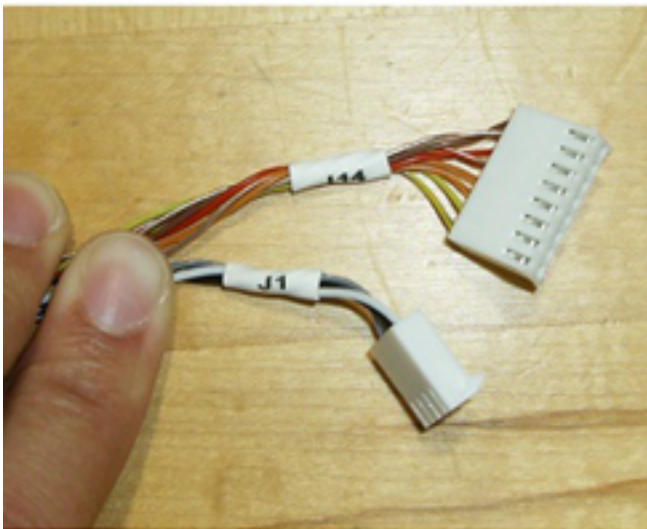
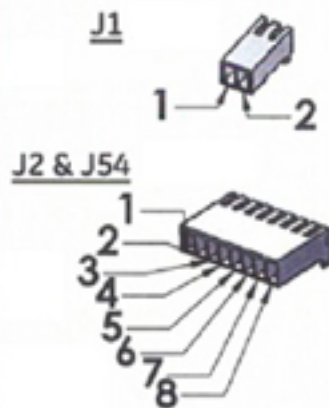
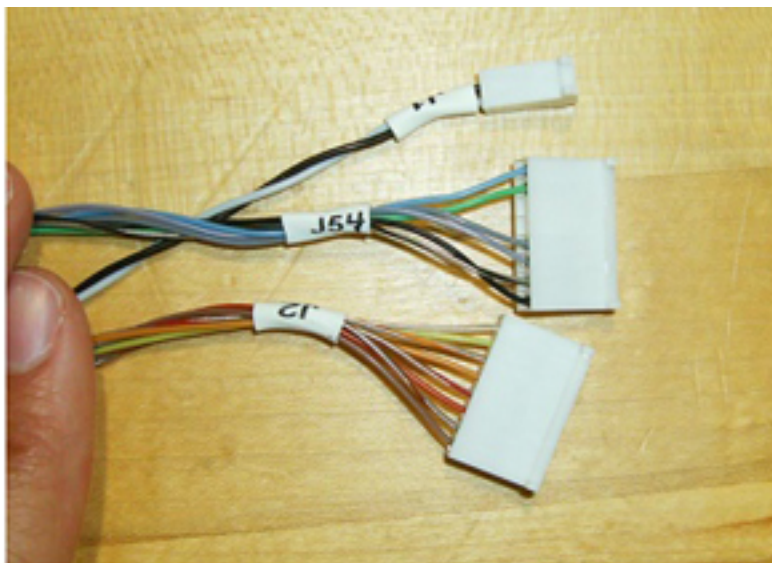


Table 3-8: "B" Connector DC Allocation

System Side Connector DC Pin	Coil Side Connector DC Pin
A3	J2-1
A4	J2-2
A5	J2-3
A6	J2-4
A7	J2-5
A8	J2-6
A9	J2-7
A10	J2-8
A1	J54-1
A2	J54-2
M7	J54-3
M8	J54-4
M2	J54-7
M5	J54-8
M3	J1-1
M6	J1-2
Short J9 to J10	N/A
Short M9 to M10	N/A

Illustration 3-11: "B" connector DC Cables From Coils Side



With the coil cable disconnected from the coil assembly, use a multimeter to ensure no open circuit condition exists between the RF ground shields of the RF coax connector in the system side and the coils side. Refer to [Table 3-9](#) and [Table 3-10](#) and [Illustration 3-6](#) and [Illustration 3-12](#) for 8Ch “B” Hypertronics connector cable and 8Ch Bendix (Legacy) connector cable. For the Bendix connector cable, use any ground pin in [Table 3-10](#) for the RF ground of coil side. Also use the multimeter to assure an open circuit condition exists between the center pin and the shield for each RF connector in the coil side.

Table 3-9: 8Ch “B” Connector RF Allocation

System Side Connector RF GND Shield	Coil Side Connector RF Shield
C1	CH1
C2	CH2
C3	CH3
C4	CH4
D1	CH5
D2	CH6
D3	CH7
D4	CH8

Illustration 3-14: 8Ch “B” Connector RF Cables from the Coils Side

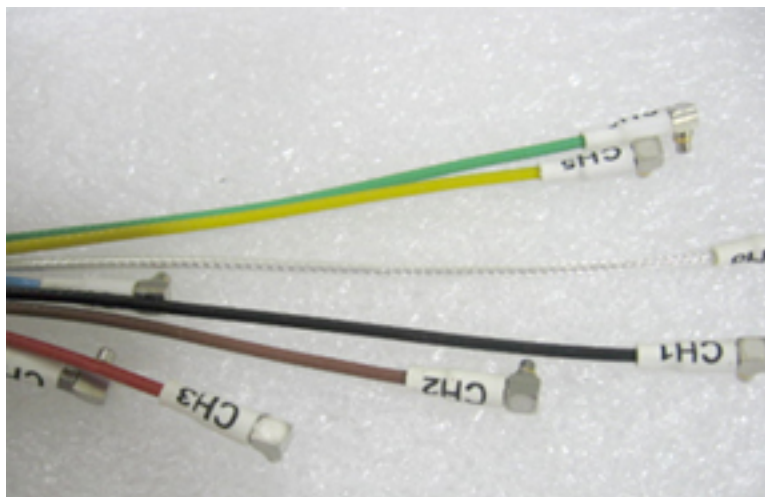
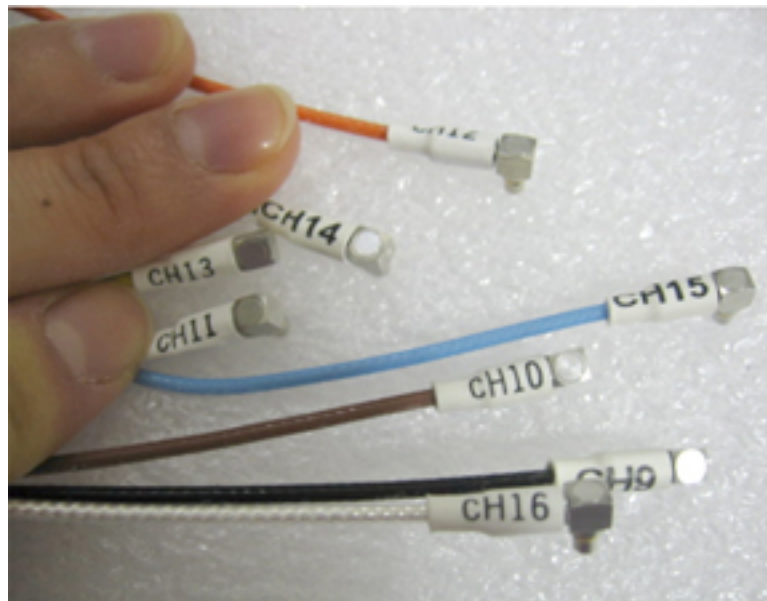


Table 3-10: 8Ch Bendix Connector RF Allocation

System Side Connector RF GND Pin	Coil Side Connector RF Shield
All following pins are grounded together. Using any of the pins is sufficient for this test. A12, A14, A16, A18, A20, A22, A24, A26, B2, B4	CH9
	CH10
	CH11
	CH12
	CH13
	CH14
	CH15
	CH16

Illustration 3-15: 8Ch Bendix Connector RF Cables from the Coils Side



If one or more of the readings indicate an open, replace the cable. Otherwise, proceed to the DC Lines Continuity Check.

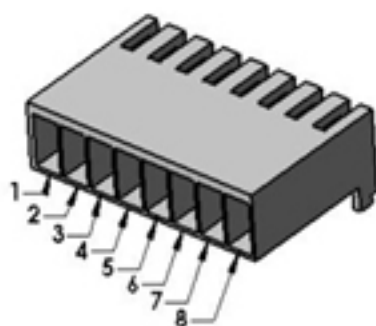
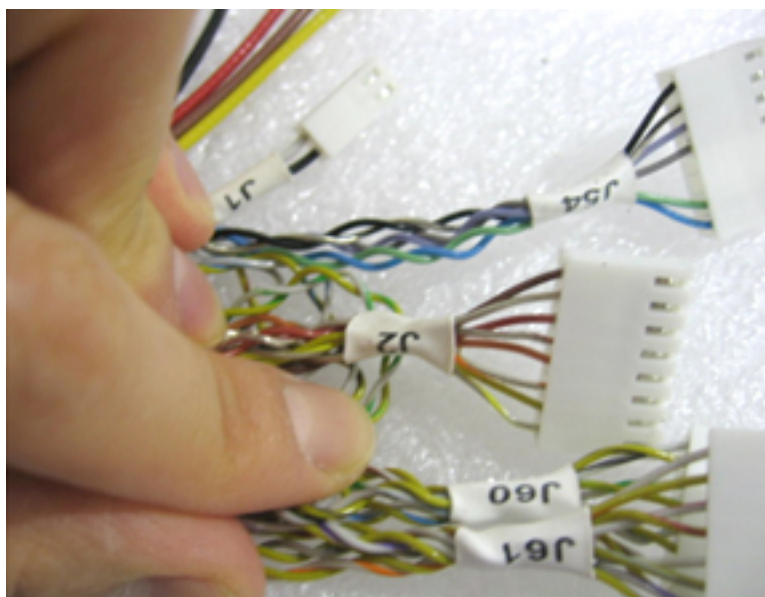
DC Lines Continuity Check

With the coil cable disconnected, use a multimeter to determine continuity between the designated pins for each control line below. Also test the continuity between the pairs of pins within system connector that are listed in the tables below. If any line or pair of pins indicates an open, replace the output cable. Please note that the DC continuity test does not apply to Bendix cable.

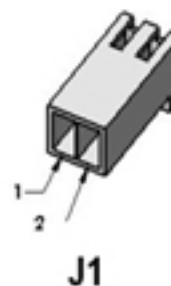
Table 3-11: "B" Connector DC Allocation for Upgrade System

System Side Connector DC Pin	Coil Side Connector DC Pin
A1	J54-1
A2	J54-2
A3	J2-1
A4	J2-2
A5	J2-3
A6	J2-4
A7	J2-5
A8	J2-6
A9	J2-7
A10	J2-8
K1	J60-2
K2	J60-8
K3	J61-4
K4	J61-2
K5	J61-8
K6	J60-4
K7	J60-6
K8	J61-6
M2	J54-7
M3	J1-1
M5	J54-8
M6	J1-2
M7	J54-3
M8	J54-4
J1	J60-1
J2	J60-7
J3	J61-3
J4	J61-1
J5	J61-7
J6	J60-3
J7	J60-5
J8	J61-5
Short J9 to J10	N/A
Short M9 to M10	N/A

Illustration 3-16: "B" Connector DC Cables from the Coils Side



J2, J54, J60, J61



J1

1.5.4.4 Upgrade System HNS Rev 3

For upgrade systems, we use 8Ch “A” Hypertronics connector cable for TL unit and 8Ch “B” Hypertronics connector cable and 8Ch Bendix (Legacy) connector for HN unit.

The upgrade Rev 3 coil uses the same 8Ch “A” Hypertronics connector cable for TL unit as the forward production Rev 3 coil. Refer to [Section 1.5.4.2](#) for checking the connector A cable.

Illustration 3-17: Pin Layouts of 8Ch “B” Hypertronics Connector for Upgrade System

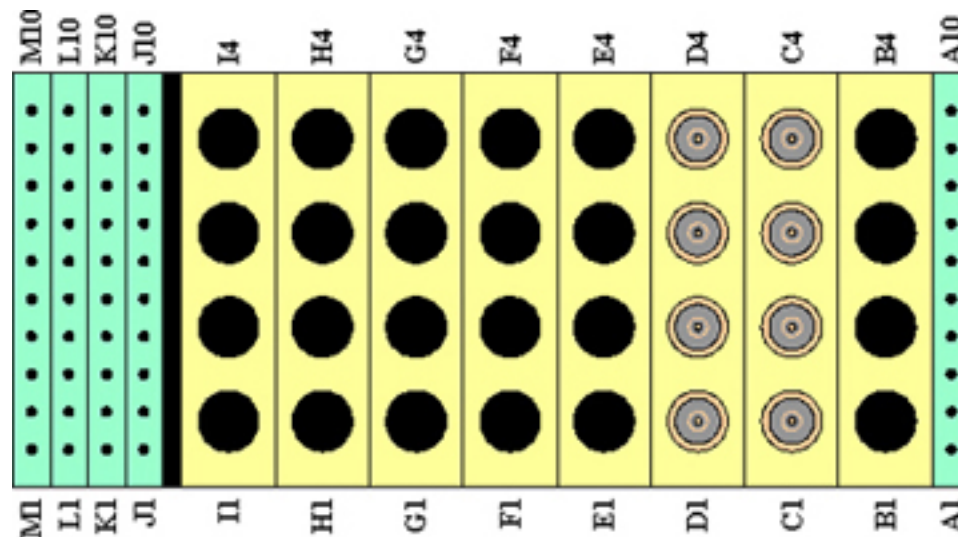
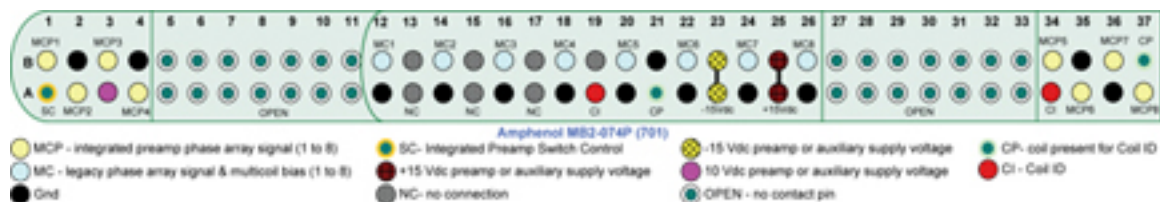


Illustration 3-18: Pin Layouts of 8Ch Bendix Connector for Upgrade System



RF Signal Channels Check

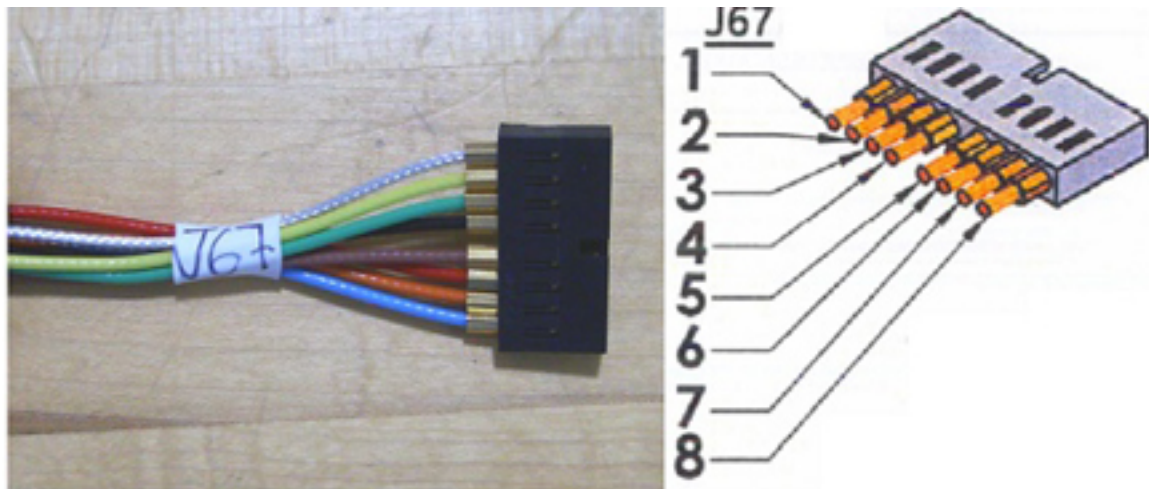
NOTE: The center pins of the System Connector receive coaxial connectors are extremely fragile. Do not attempt to make any measurements at these center pins.

With the “B” Connector cable disconnected from the coil assembly, use a multimeter to ensure a short circuit condition exists between the designated RF GND shields detailed in the table below. Refer to [Table 3-12](#) and [Illustration 3-17](#) and [Illustration 3-19](#).

Table 3-12: 8Ch "B" Connector RF Allocation

C1	4
C2	5
C3	6
C4	7
D1	2
D2	3
D3	8
D4	1

Illustration 3-19: 8Ch "B" Connector RF Cables From The Coils Side

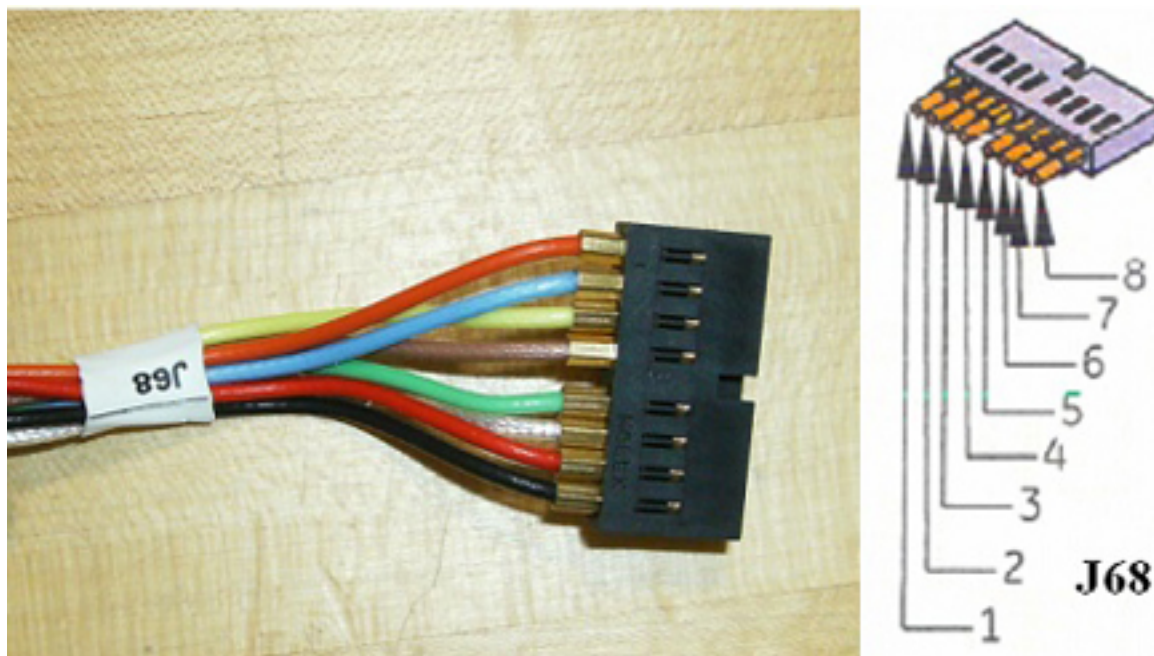


With the Bendix Connector cable disconnected from the coil assembly, use a multimeter to ensure a short circuit condition exists between the designated GND shields. Refer to [Table 3-13](#) and [Illustration 3-17](#) and [Illustration 3-20](#). Use any ground pin in Table 13 for the RF ground of coil side.

Table 3-13: 8Ch Bendix Connector RF Allocation

System Side Connector RF GND Shield	Coil Side Connector RF Shield (All located on Gang J68)
All following pins are grounded together. Using any of the pins is sufficient for this test. A12, A14, A16, A18, A20, A22, A24, A26, B2, B4	1
	2
	3
	4
	5
	6
	7
	8

Illustration 3-20: 8Ch Bendix Connector RF Cables From The Coils Side



If one or more of the readings indicate an open, replace the cable. Otherwise, proceed to the DC Lines Continuity Check.

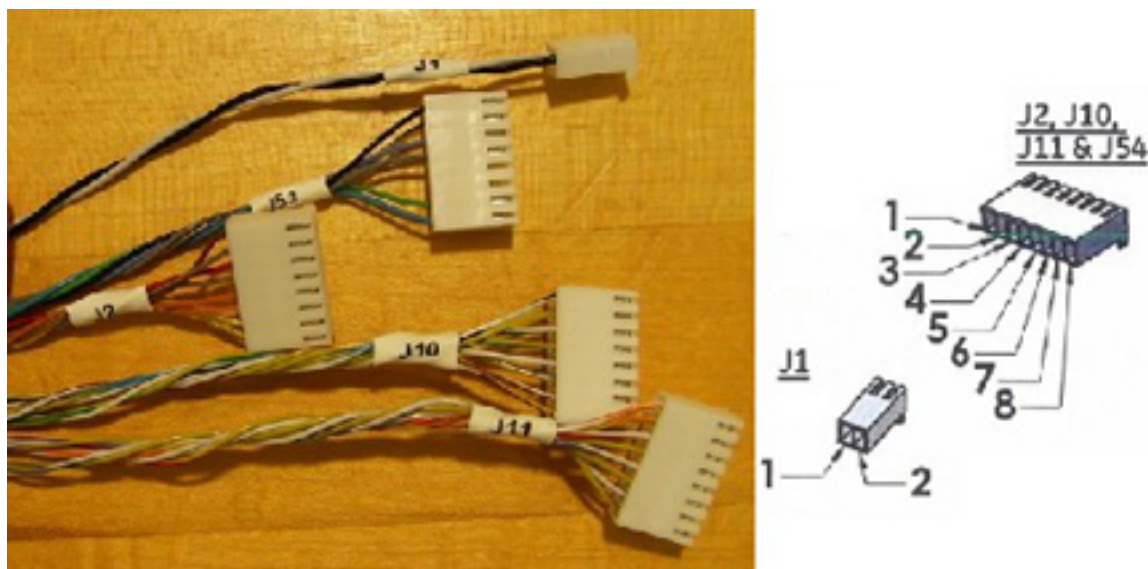
DC Lines Continuity Check

With the coil cable disconnected, use a multimeter to determine continuity between the designated pins for each DC control line below. Also test the continuity between the pairs of pins within system connector that are listed in the tables below. If any line or pair of pins indicates an open, replace the output cable. Please note that this DC continuity test does not apply to Bendix connector.

Table 3-14: "B" Connector DC Allocation For Upgrade System

System Side Connector DC Pin	Coil Side Connector DC Pin
A1	J54-1
A2	J54-2
A3	J2-1
A4	J2-2
A5	J2-3
A6	J2-4
A7	J2-5
A8	J2-6
A9	J2-7
A10	J2-8
J1	J10-1
J2	J10-7
J3	J11-3
J4	J11-1
J5	J11-7
J6	J10-3
J7	J10-5
J8	J11-5
K1	J10-2
K2	J10-8
K3	J11-4
K4	J11-2
K5	J11-8
K6	J10-4
K7	J10-6
K8	J11-6
M2	J54-7
M3	J1-1
M5	J54-8
M6	J1-2
M7	J54-3
M8	J54-4
Short J9 to J10	N/A
Short M9 to M10	N/A

Illustration 3-21: "B" Connector DC Cables from the Coils Side



1.5.4.5 Discovery MR450 System HNS Rev 3

8ch "A" Hypertonic connector cable is attached to the TL unit and 16ch "P" P-connector plug cable is attached to HNU on the Discovery system.

Illustration 3-22: Pin Layout of "P" P-connector

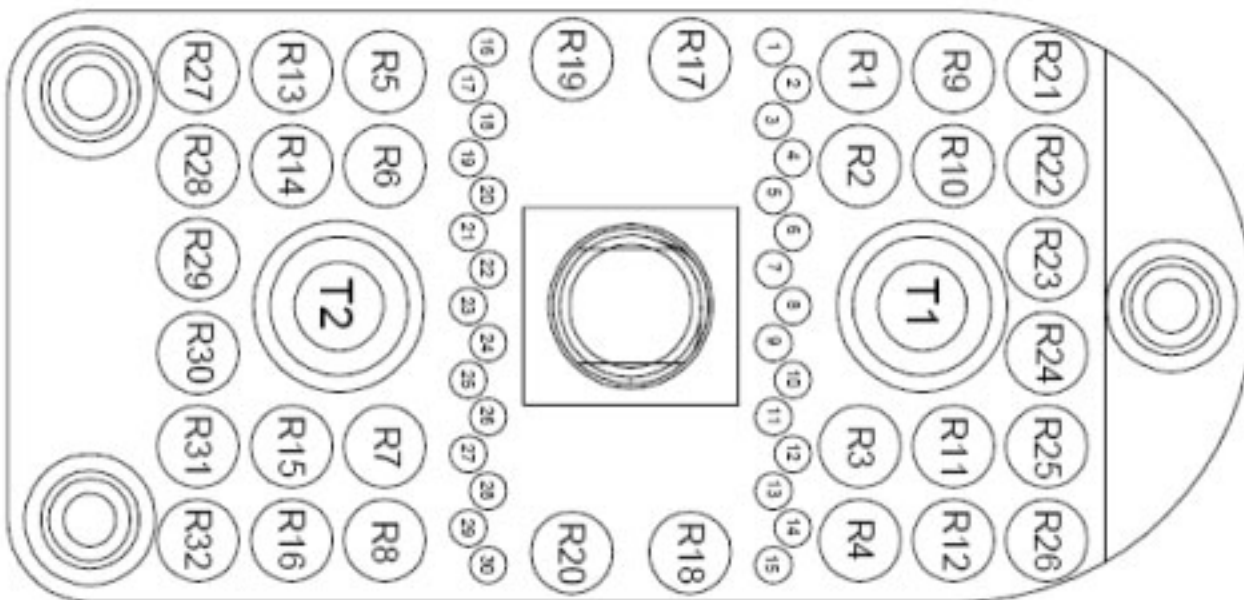
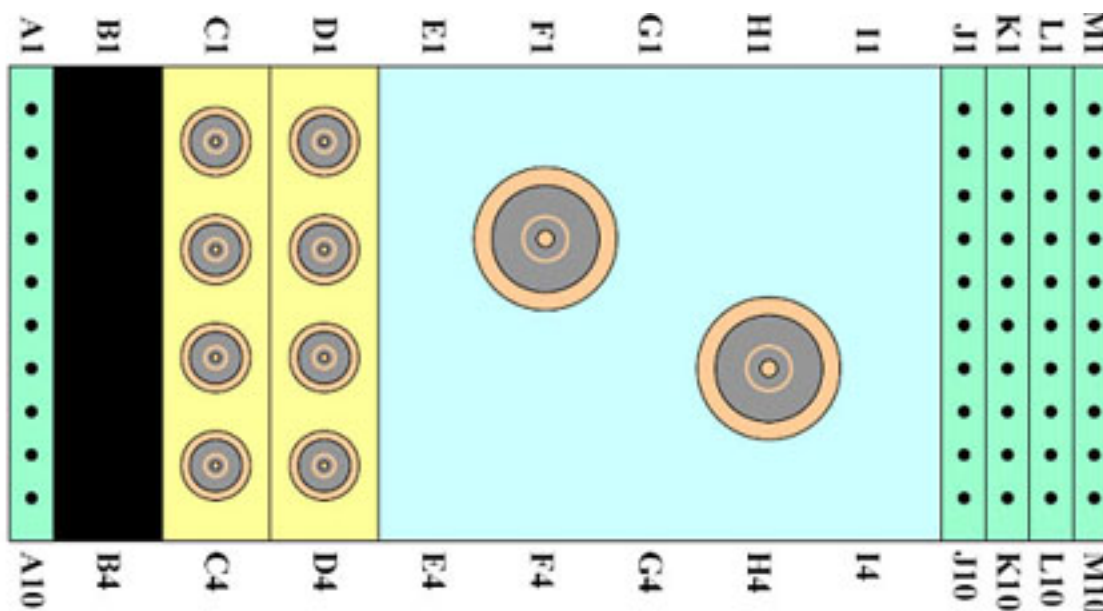


Illustration 3-23: "A" Hypertronics connector



NOTE: The center pins of the system connector that receive coaxial connectors are extremely fragile. Do not attempt to make any measurements at these center pins.

The following procedure will ensure the integrity of the ground section for each RF coax cable, which are detailed in the following tables.

1. With the coil cable disconnected, use a multimeter to ensure an open circuit condition exists between the RF ground shields of the RF coax connector in the system side and the coils side.
2. If one or more of the readings indicate an "open", replace the cable. If no "open" is found for the ground of the RF cables, proceed to the DC Lines Continuity Check.

Table 3-15: 8ch "A" Connector RF Allocation

System Side Connector	Coil Side Connector RF GND Shield Pin#
RF GND Shield	(All located on Gang Connector J28)
C1	8
C2	1
C3	7
C4	2
D1	6
D2	3
D3	5
D4	4

Illustration 3-24: 8ch "A" Connector RF Cables from Coil Side

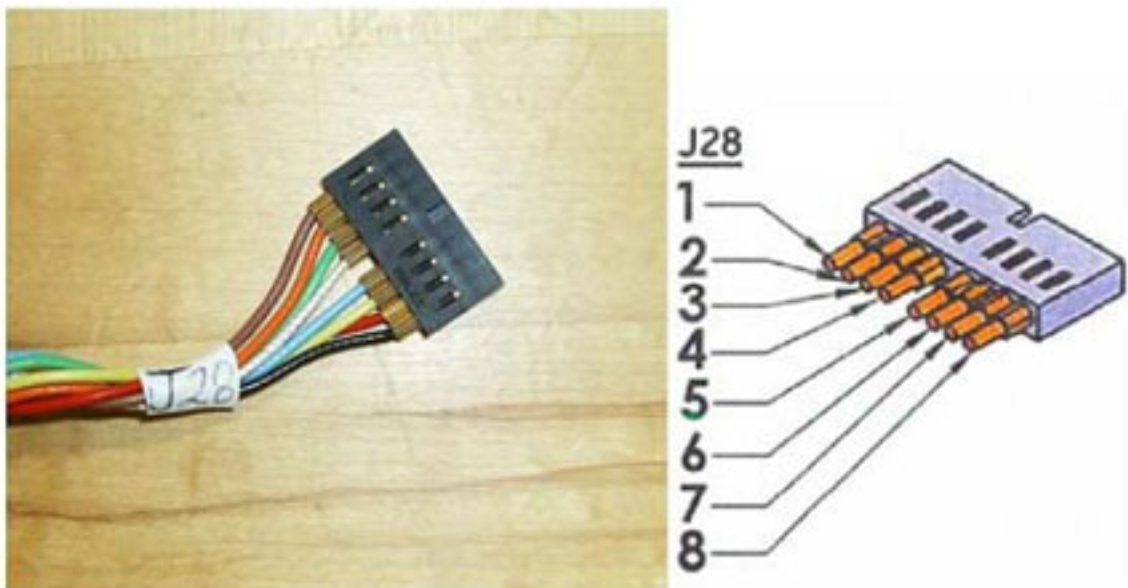
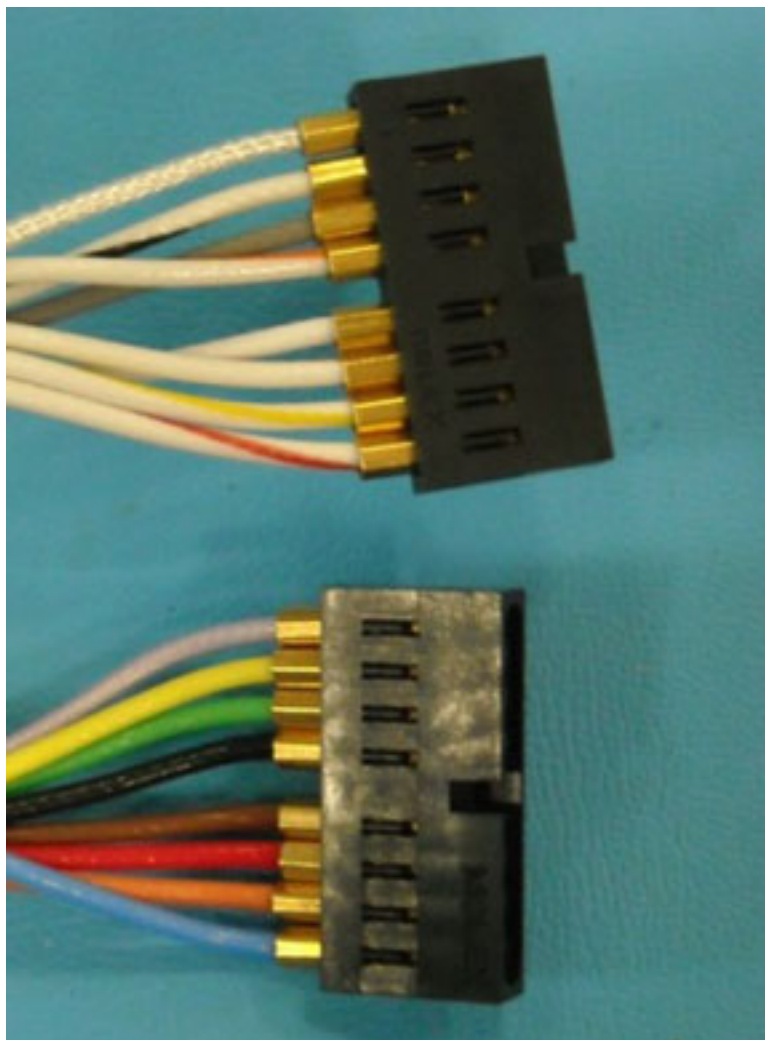


Table 3-16: 16ch “P” P-connector RF Allocation

System Side Connector RF GND Shield	Coil Side Connector RF GND Shield Pin# (All located on the Gang Con- nector J67)	System Side Connector RF GND Shield	Coil Side Connector RF GND Shield Pin# (All located on connector J68)
R1	4	R9	3
R2	5	R10	5
R3	6	R11	2
R4	7	R12	6
R5	2	R13	8
R6	3	R14	4
R7	8	R15	7
R8	1	R16	1

Illustration 3-25: 16ch “P” Connector RF Cables from Coil Side



DC Lines Continuity Check

With the coil cable disconnected, use a multimeter to determine continuity between the designated Pins for each DC control line, as indicated below. Also, test the continuity between the pairs of Pins within the system connector that are listed in the table below. If any line or pair of Pins indicates an “open”, replace the output cable.

Table 3-17: “A” Connector DC Allocation

System Side Connector	Coil Side Connector
DC Pin	DC Pin
M3	J1-1
M6	J1-2
M8	J1-3
M7	J1-4
J1	J14-4
K1	J14-2
J2	J14-3
K2	J14-4
J3	J14-5
K3	J14-6
J4	J14-7
K4	J14-8
Short A1 to A2	NA
Short J9 to J10	NA
Short M9 to M10	NA

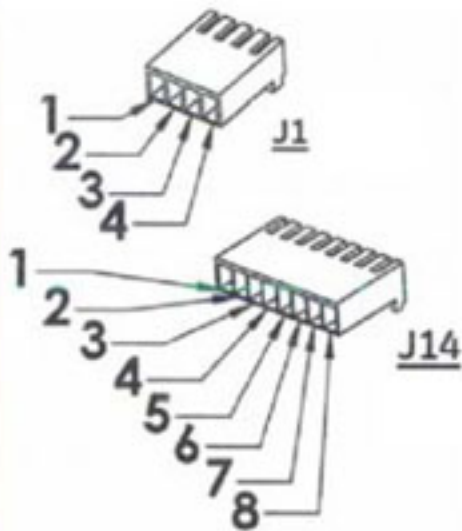
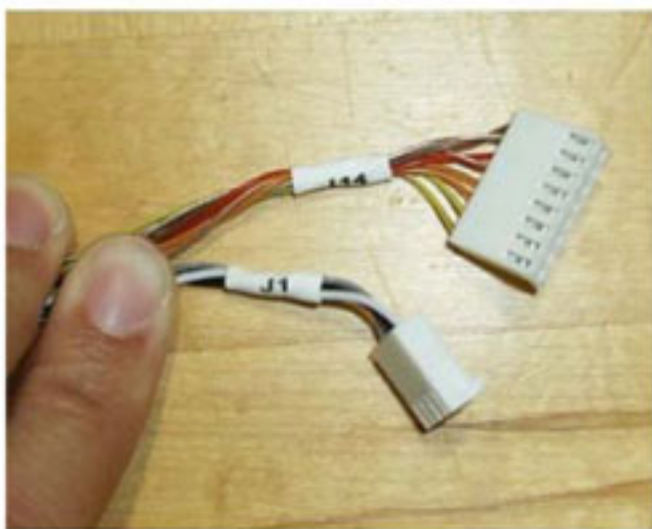
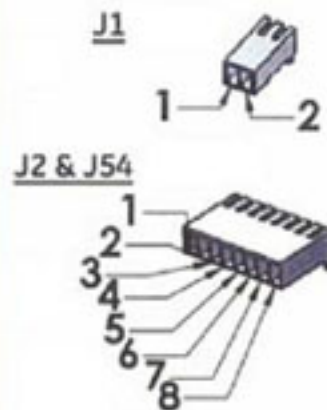
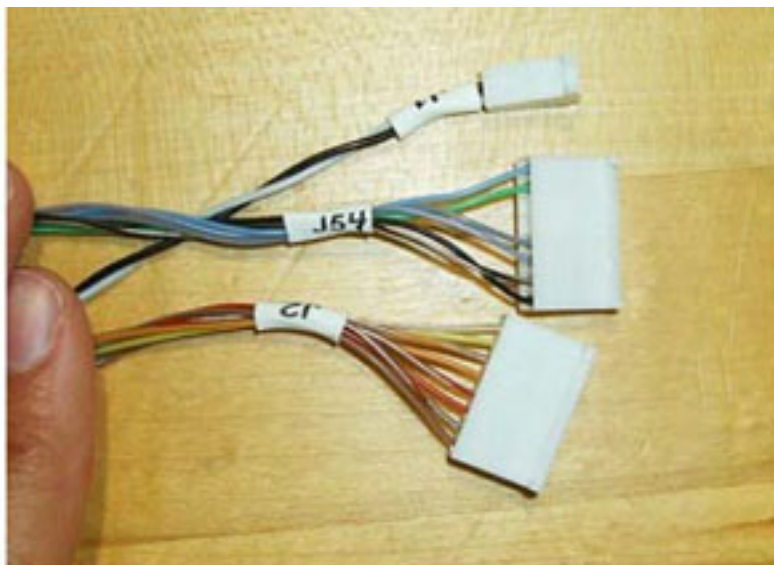
Illustration 3-26: “A” connector DC Cables from Coils Side

Table 3-18: "P" P-connector DC Allocation

System Side Connector	Coil Side Connector
DC Pin	DC Pin
7	J2-1
8	J2-2
9	J2-3
10	J2-4
11	J2-5
12	J2-6
13	J2-7
14	J2-8
1	J54-1
2	J54-2
5	J54-3
6	J54-4
20	J54-7
21	J54-8
18	J1-1
19	J1-2
Short 4 to 3	NA

Illustration 3-27: "P" P-connector DC Cables from Coil Side



1.5.4.6 Configuration of Jumpers on the Interface Board for Upgrade and Forward Production System

When you replace the cable assembly, you need to make sure the configuration of jumpers on the interface board for upgrade and forward production systems.

Illustration 3-28: Interface Board for HNS Rev 2 Coil



Table 3-19: Configuration of Jumpers on the HNS Rev 2 Interface Board

	Upgrade System	Forward Production HDx System	Discovery System
Jumpers	Put the jumpers JMPR4, 5, 6, 7, 8, 9, 10	None	None

Illustration 3-29: Interface Board for HNS Rev 3 Coil

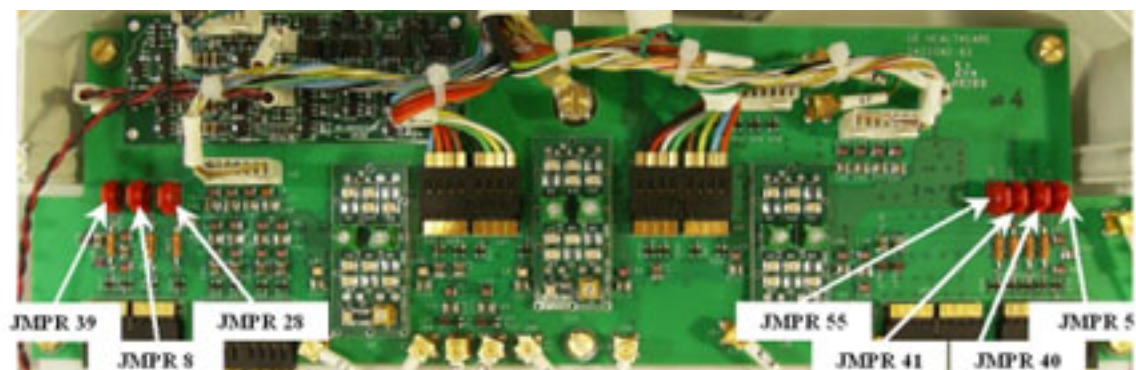


Table 3-20: Configuration of Jumpers on the HNS Rev 3 Interface Board

	Upgrade HDx System	Forward Production HDx System	Discovery System
Jumpers	Put the jumpers JMPR 5, 8, 28, 39, 40, 41, 55	None	None

* *The part number of Jumper for shorting is U1-410002.

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Chapter 4 Parts

1 1.5T Head Neck Spine (HNS) Coil Field Replacement Units

1.1 HNS Rev 2 Coil FRUs

Table 4-1: 1.5T HNS FRUs

Part No.	Description
2417362	Head Neck Unit (A&P)
2418074	HNU Adapter Block
2417360	Neck Chest Unit
2417361	TL Unit
2417356	Upgrade Output Cable-Hypertronics
2417357	Upgrade Output Cable (Bendix)
2417359	Forward Production Output Cable
2417358	TL Unit Cable
2418080	Mirror

1.2 HNS Rev 2 Coil Accessories

Table 4-2: 1.5T HNS Accessories

Part No.	Description
2418076	TL Phantom Kit
2418078	HN Phantom Kit
2418081	Pad Kit

1.3 HNS Rev 3 Coil HDx and Discovery 450 System FRUs

Table 4-3: 1.5T HNS FRUs

Part No.	Description
2423192*	Head Neck Unit 1.5T Brain Spine
2423193	Neck Chest Unit 1.5T Brain Spine
2423194	TL Unit 1.5T Brain Spine
2423319	8 Ch HNS EXC III Upgrade System Cable
2423317	8 Ch EXC II HNS Forward Production System Cable
2423320	16 Ch EXC III HNS Forward Production System Cable
2424269	16 Ch HNS P-Connector System Cable FRU to the Table
2423318	HNS TL Unit Cable
2419521	Mirror Assembly
* Head Neck Unit FRU includes the Anterior, Posterior & Adapter Block sections	

1.4 HNS Rev 3 HDx and Discovery 450 System FRUs

Table 4-4: 1.5T HNS Accessories

Part No.	Description
2418076	TL Phantom Kit
2418078	HN Phantom Kit
2418081	Pad Kit

Chapter 5 Replacements

1 1.5T Head Neck Spine (HNS) Coil Cable Replacement

1.1 Personnel Requirements

Personnel Requirements	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	40 mins	Not Applicable

1.2 Overview

Follow this process to change the system cable for coils that have a cable available as a spare part. This process will be the same whether replacing the coil or the cable.

1.3 Preliminary Requirements

1.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Standard tool kit – Flathead screwdriver and pliers	1	-	-	-

1.3.2 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Output Cable-Hypertronics (8 Channel “B” connector) for the Head Neck section with the coil id board	1	HNS Rev. 2 Upgrade with the coil id board	2417356	-
Output Cable-bendix (without the coil id board)	1	HNS Rev. 2 Upgrade without the coil id board	2417357	-
Output Cable (16 Channel “B” connector) for the Head Neck Section with Coil id board.	1	HNS Rev. 2 Forward Production with Coil id board	2417359	-
TL Unit Cable	1	HNS Rev. 2 without the coil id board	2417358	-
8 CH HNS EXC III UPGRD SYS CABLE FRU	1	HNS Rev. 3 Upgrade with the coil id board	2423319	-

Item	Qty	Effectivity	Part#	Manufacturer
8 CH HNS EXC II SYS CABLE FRU	1	HNS Rev. 3 Upgrade without the coil id board	2423317	-
16 CH EXC III HNS FWD PRD SYS CBL FRU	1	HNS Rev. 3 Forward Production with Coil id board	2423320	-
HNS TL UNIT CABLE FRU	1	HNS Rev. 3 without the coil id board	2423318	-
16ch P-connector HNS Discovery System Cable FRU	1	HNS P- connector Discovery with coil id board	2424269	-

1.4 Procedure

1.4.1 HNS Rev. 2



NOTICE

Please refer to the cable FRU list above in Replacement Parts.

- If the cable FRU is with the coil id board, then the coil id board has to be removed as well.
- If the cable FRU is without the coil id board, then do not remove the coil id board.

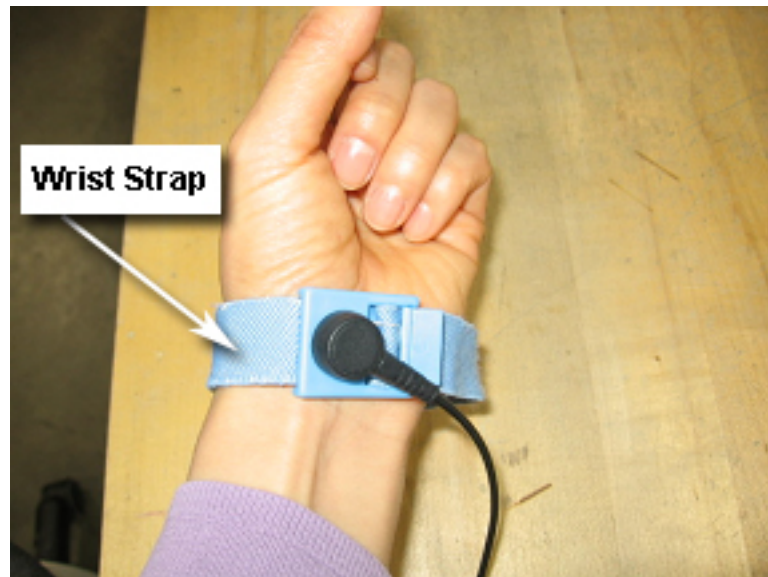


NOTICE

Observe anti-static precautions: Use wrist strap and ESD workstation (ESD mat connected to ground and place to connect ESD wrist strap).

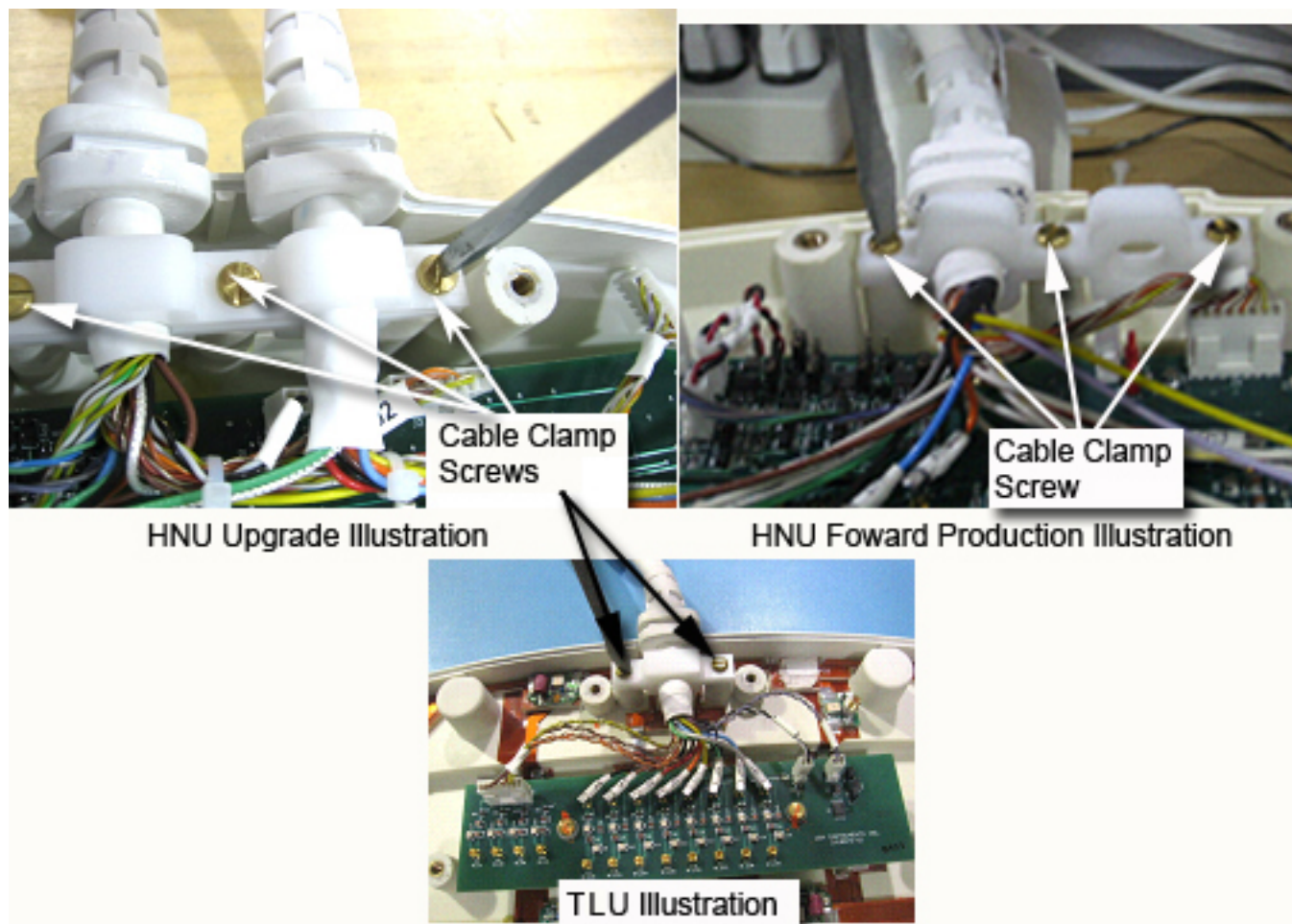
1. Put an ESD wrist strap on and ground.

Illustration 5-1: ESD Wrist Strap On



2. Unscrew the bottom cover and remove from coil. Retain the original screws used.
3. Unscrew the cable clamp from the coil as shown in [Illustration 5-2](#). Retain the original screws used.

Illustration 5-2: Removing the Cable Clamp



4. Identify and remove RF connectors (*CH* in Head Neck Unit (HNU), *J* in Thoracic Lumbar Unit (TLU)) from interface Board, see [Illustration 5-3](#). Pliers may be necessary for removal of the MMCX connectors, as they may be difficult to unplug. This will help prevent accidental breakage of the cables.

Illustration 5-3: RF Connectors In the HNU

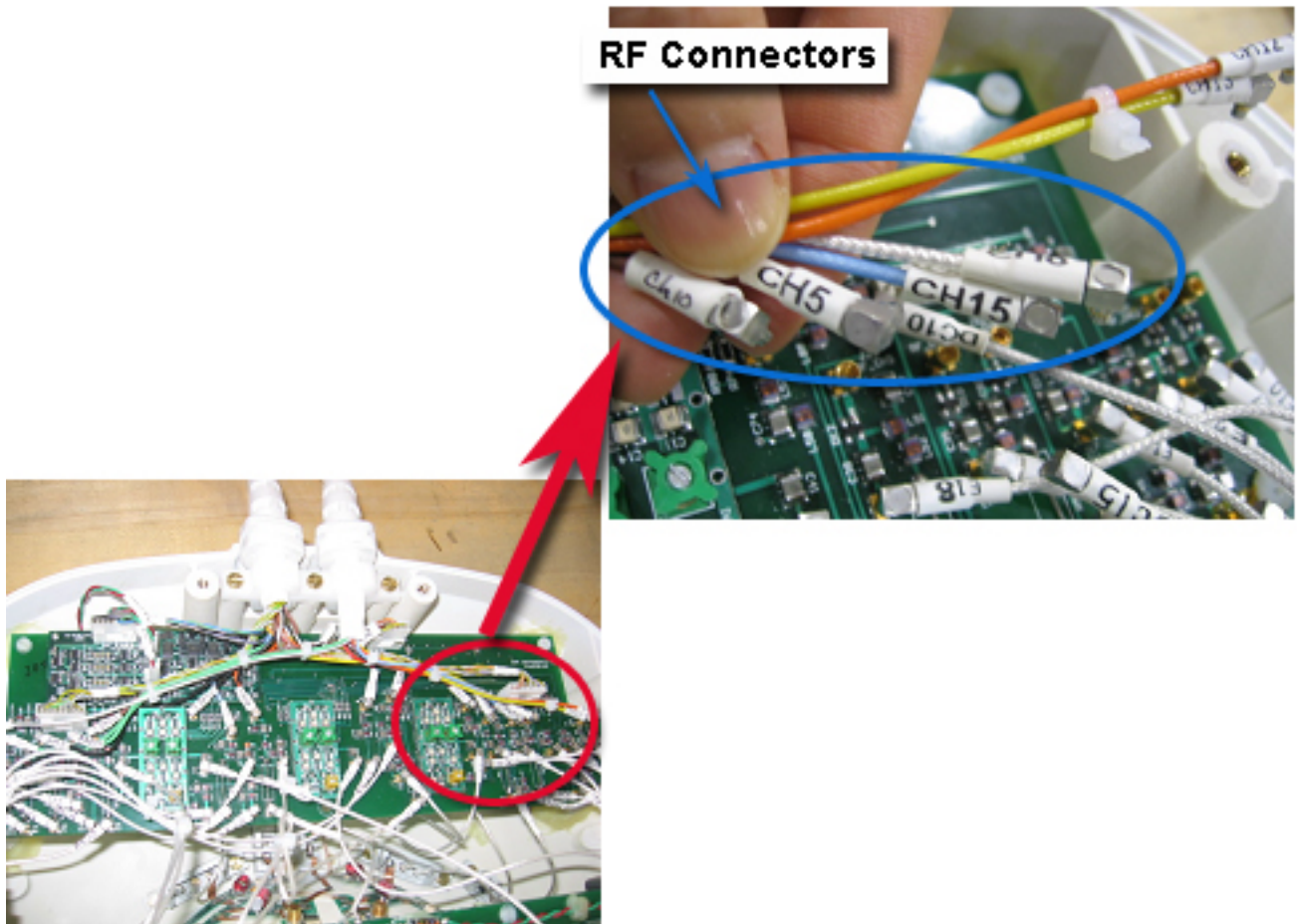


Illustration 5-4: Using Pliers For Removal of Connectors

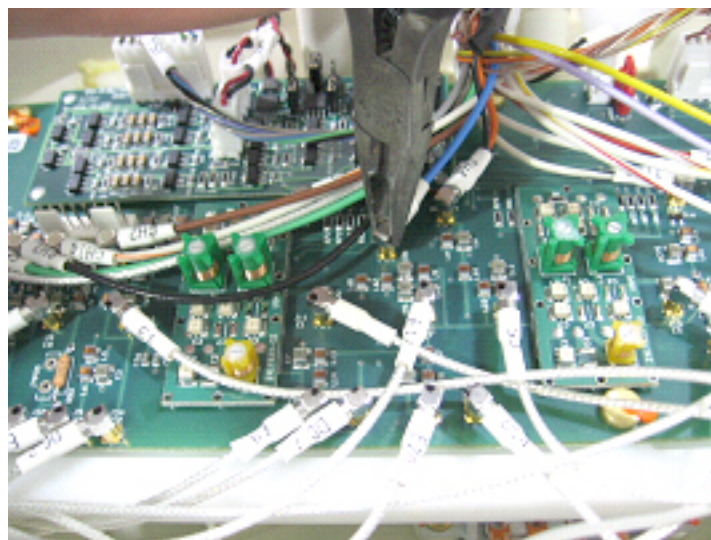
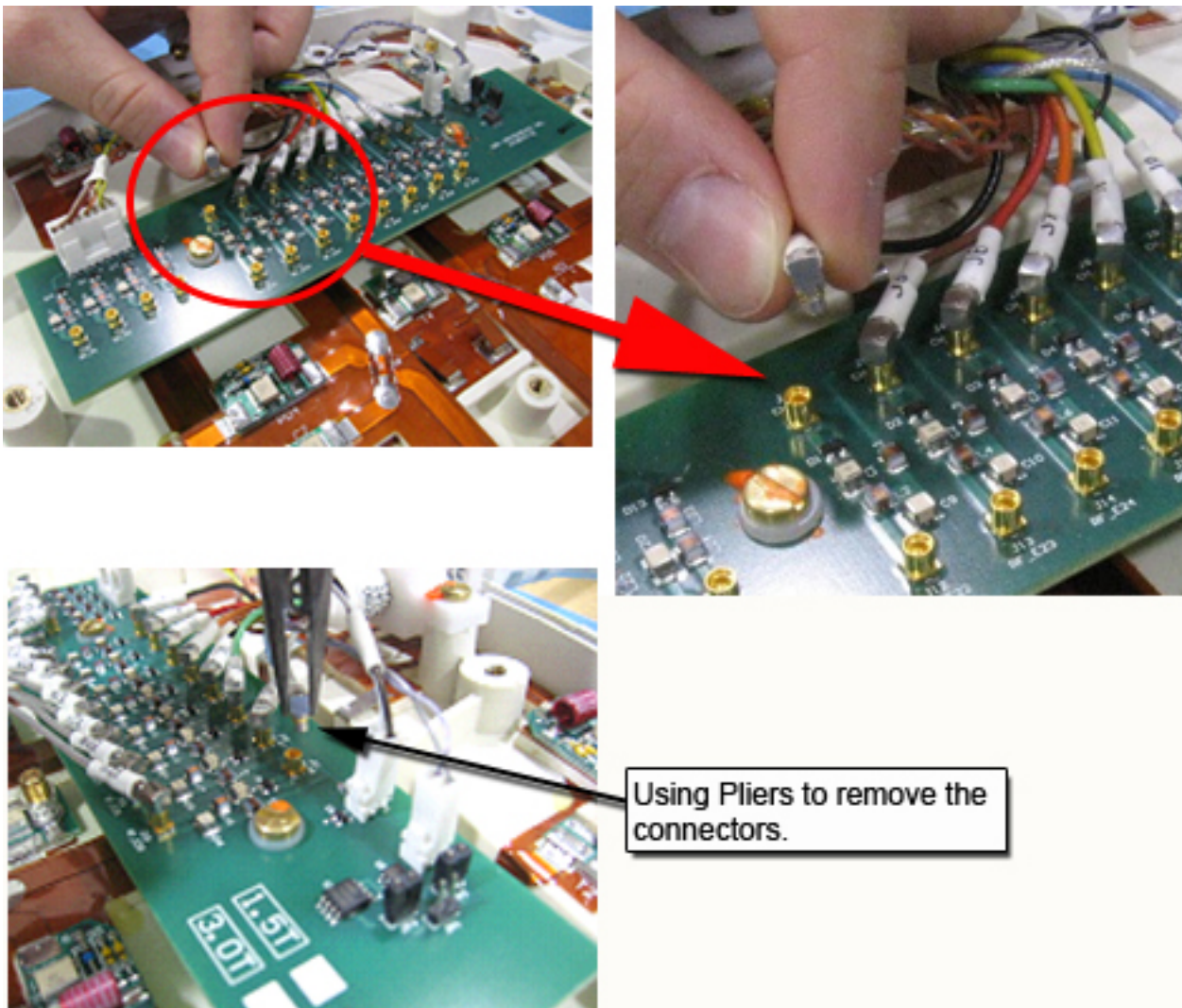


Illustration 5-5: RF Connectors for the Thoracic Lumbar Unit



5. Identify and remove DC connectors (DC) from Interface Board and Coil ID PCB. See [Illustration 5-6](#) through [Illustration 5-8](#).

Illustration 5-6: DC Connectors: Forward Production Coil (1 System Cable) Head Neck Unit (HNU)

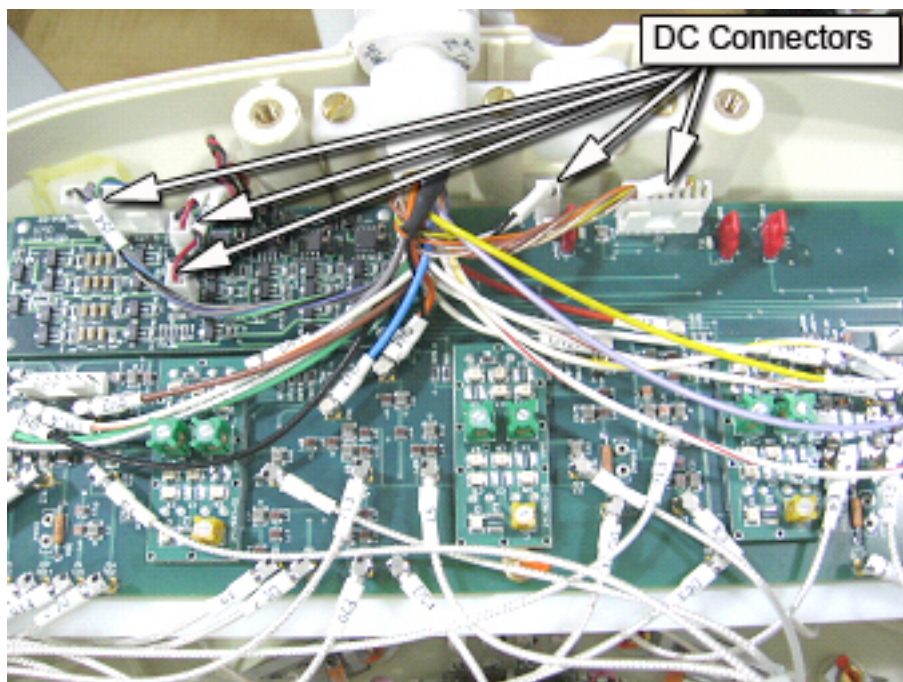


Illustration 5-7: DC Connectors: HNU Illustration (2 System Cables)(Upgrade Specific)

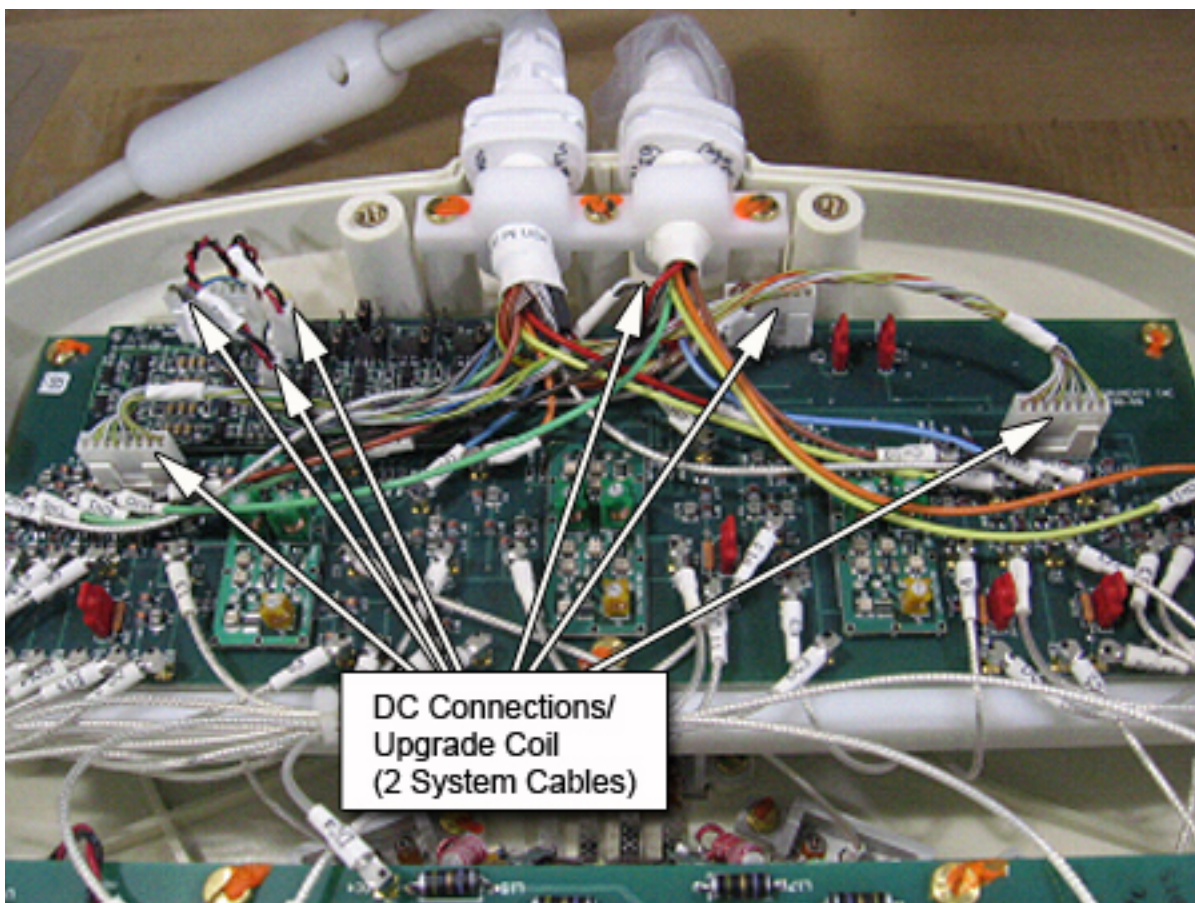
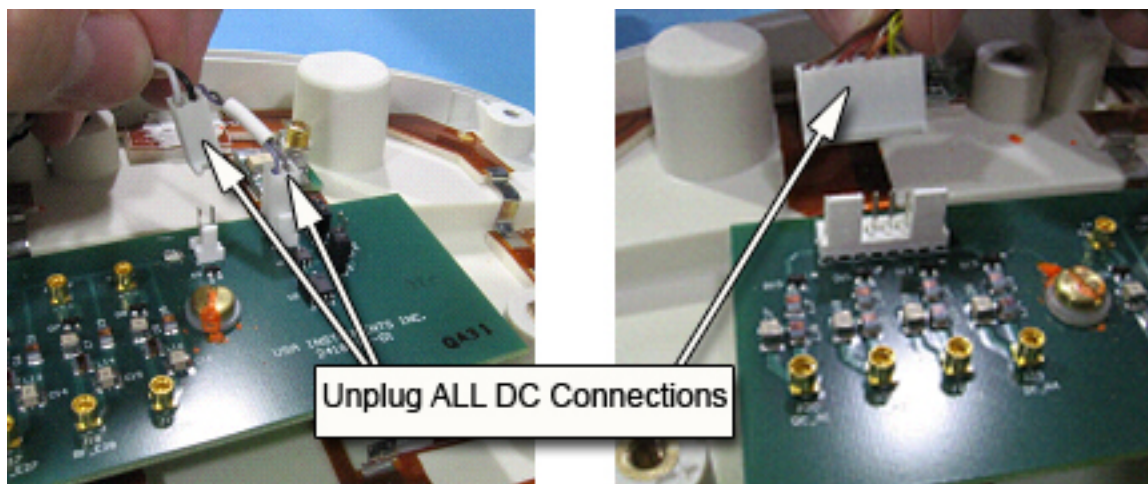


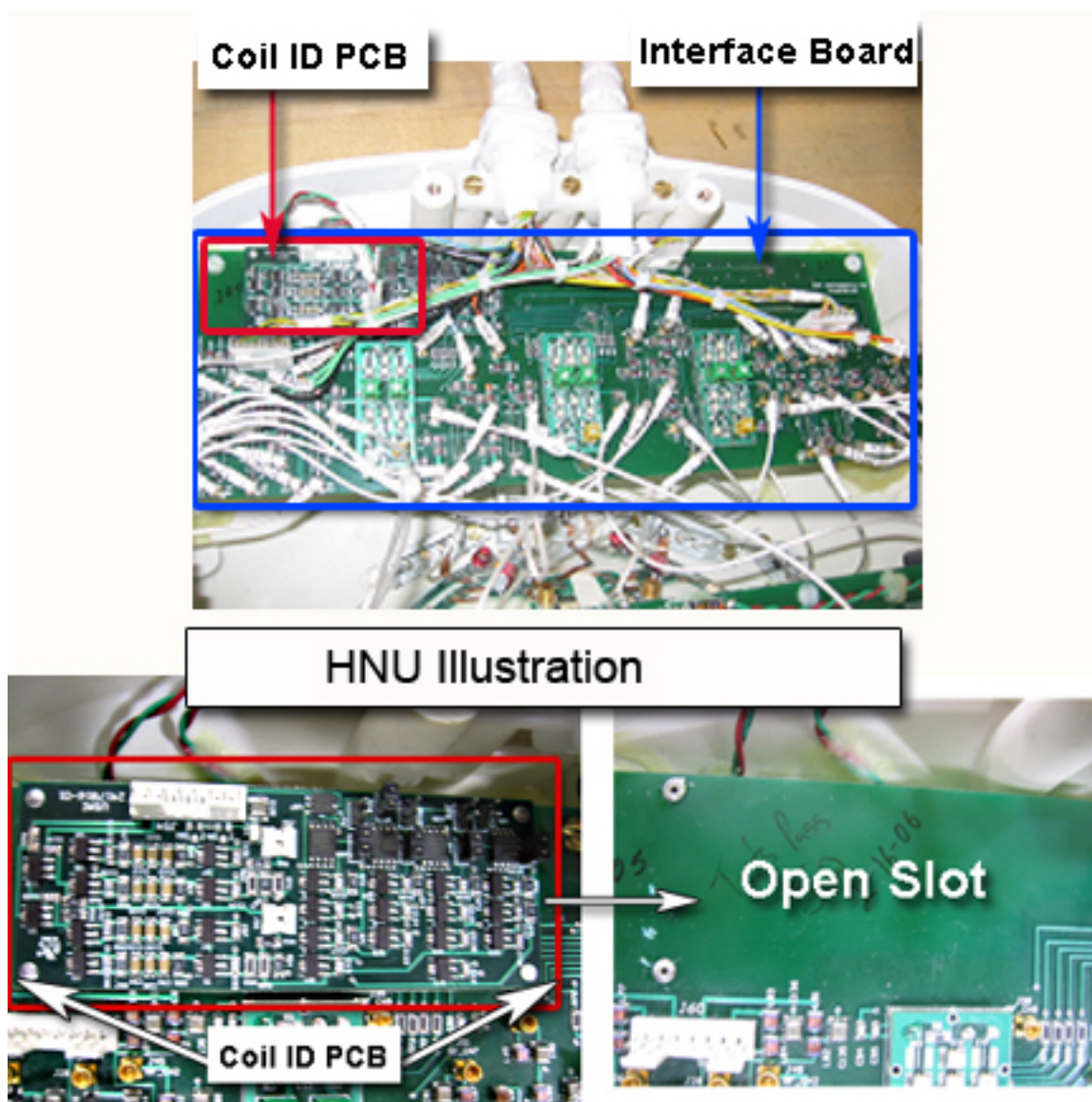
Illustration 5-8: DC Connections: TLU



6. Remove the system cable (with or without the coil id).

7. Refer to the Replacement Parts section, if the cable FRU is with the coil id; remove the coil ID PCB from Interface Board (proceed to the next step). If the cable FRU exists without the coil id board, then remove only the cable.
8. Identify the Interface Board Assembly and the coil ID PCB located on the Interface Board (Head Neck Unit (HNU) Only). Carefully remove the Coil ID assembly. See [Illustration 5-9](#).

Illustration 5-9: Interface Board and Coil ID PCB Locations



9. Install the new system cable.
10. Align the cable clamp of the cable into the notch of former slot. Screw down cable clamp using original screws. See [Illustration 5-2](#).

11. If applicable from [Step 7](#): Mount the new coil ID PCB assembly in the open slot on the Interface Board (HNU Only). See [Illustration 5-9](#).
12. Connect RF connectors to Interface Board. Verify that all labels on the connectors match those on the Interface Board. See [Illustration 5-3](#).
13. Connect DC connectors to Interface Boards and coil ID PCB (Coil ID PCB for HNU Only). DC connector "S1" connects to J55 on the coil ID PCB. DC connector "S2" connects to J56 on the coil ID PCB. Verify that all labels on the connectors match those on the Interface Board. See [Illustration 5-6](#).
14. Replace cover. Use original screws to fasten cover.

1.4.2 HNS Rev. 3 HDx and Discovery 450 Systems



NOTICE

Please refer to the cable FRU list above in Replacement Parts.

- If the cable FRU is with the coil id board, then the coil id board has to be removed as well.
- If the cable FRU is without the coil id board, then do not remove the coil id board.

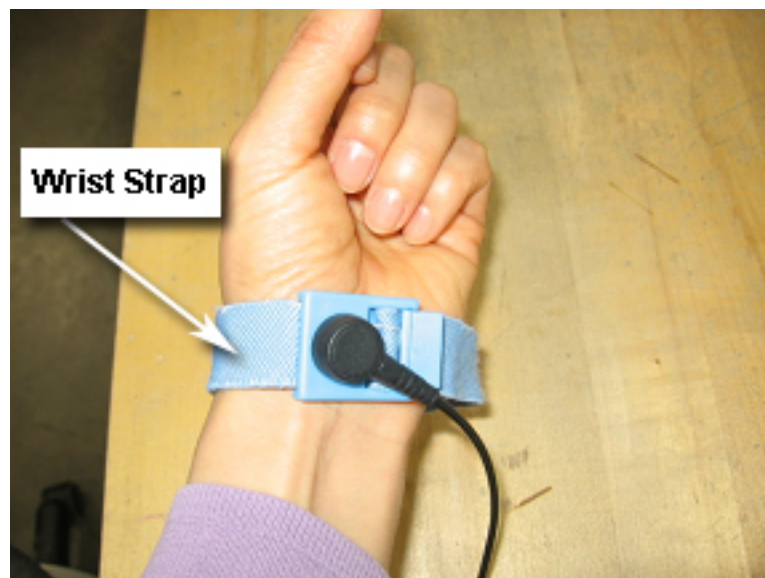


NOTICE

Observe anti-static precautions: Use wrist strap and ESD workstation (ESD mat connected to ground and place to connect ESD wrist strap).

1. Put an ESD wrist strap on and ground.

Illustration 5-10: ESD Wrist Strap On

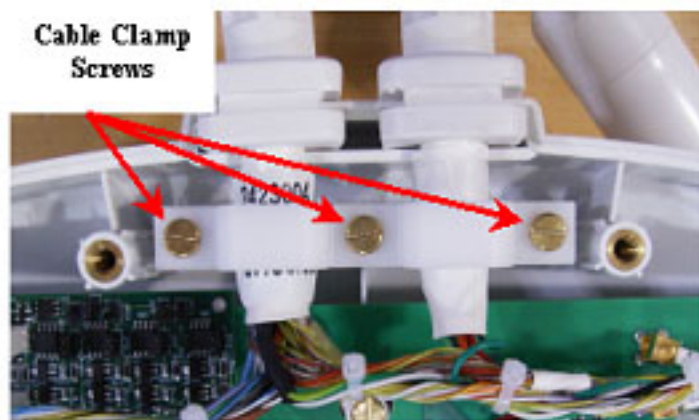


2. Unscrew the bottom cover and remove from coil. Retain the original screws used.

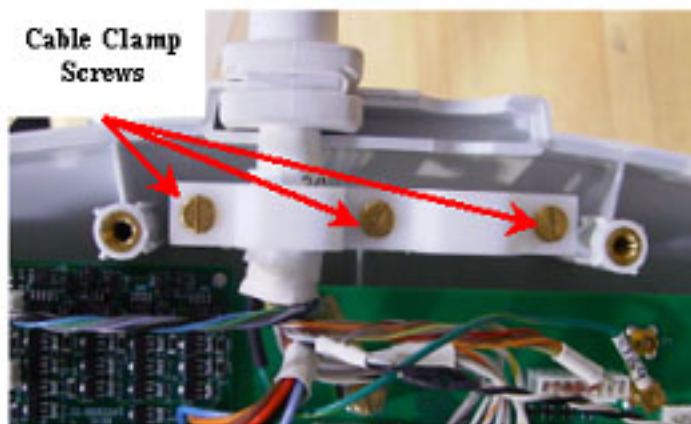
3. Unscrew the cable clamp from the coil as shown in [Illustration 5-11](#). Retain the original screws used.

NOTE: When re-installing components, tighten all screws snugly, but do not over-tighten. The recommended torque is 10 in-lbs for M4 screws and 15 in-lbs for M5 screws.

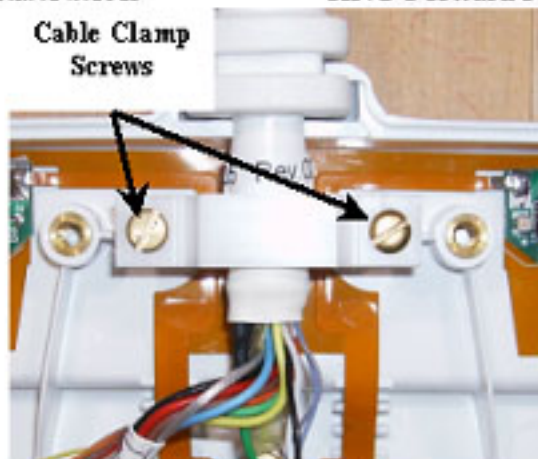
Illustration 5-11: Removing the Cable Clamp



HNU Upgrade Illustration



HNU Forward Production & Discovery Illustration



TLU Illustration

4. Identify and remove ganged RF connectors from interface Board. See [Illustration 5-12](#).

Illustration 5-12: RF Connectors in the HNU

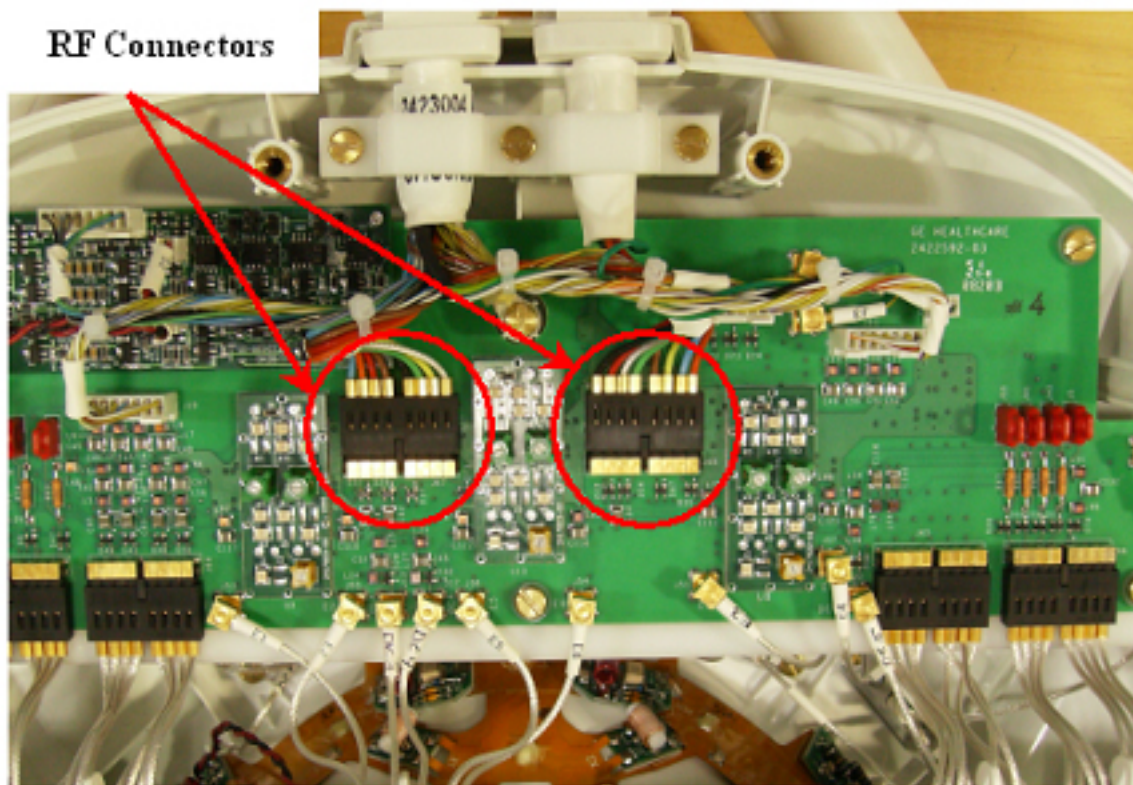
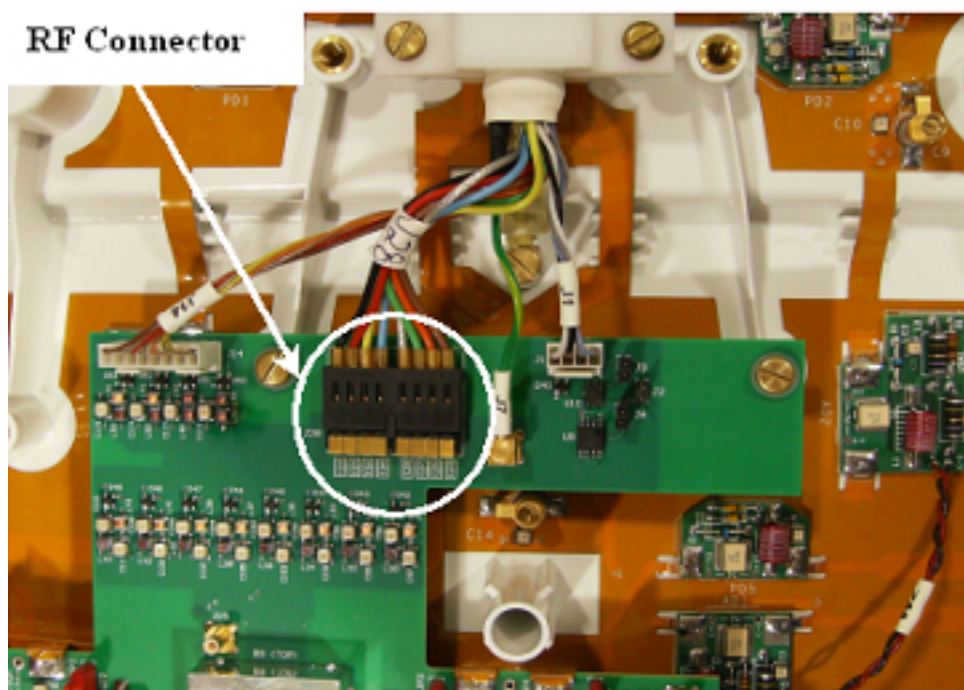


Illustration 5-13: RF Connectors for the Thoracic Lumbar Unit



5. Identify and remove DC connectors (DC) from Interface Board and Coil ID PCB. See [Illustration 5-14](#) through [Illustration 5-16](#).

Illustration 5-14: DC Connectors: Discovery and Forward Production Coil (1 System Cable) HNU

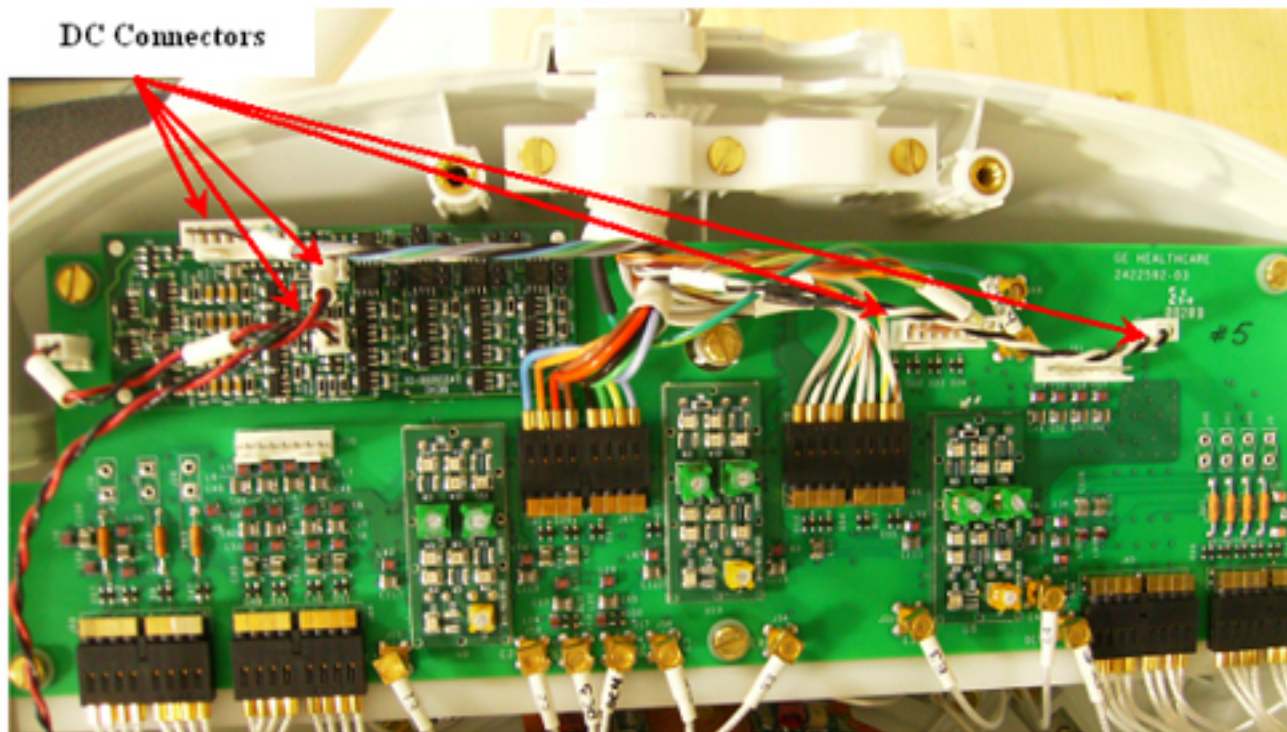


Illustration 5-15: DC Connectors: HNU Illustration (2 System Cables) (Upgrade Specific)

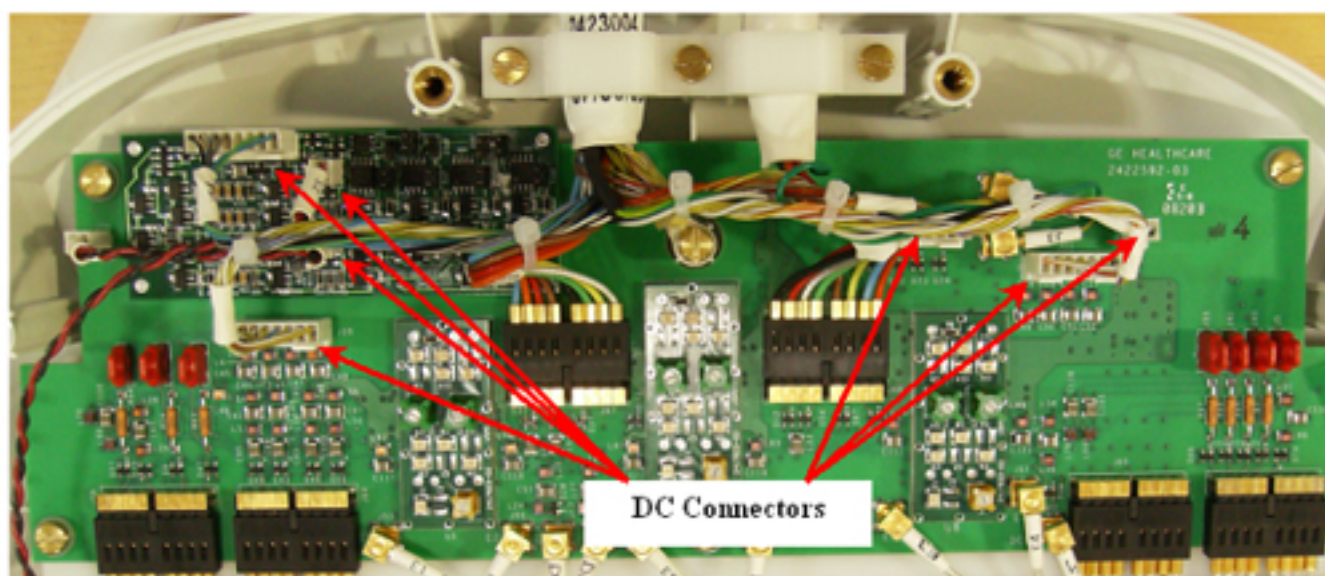
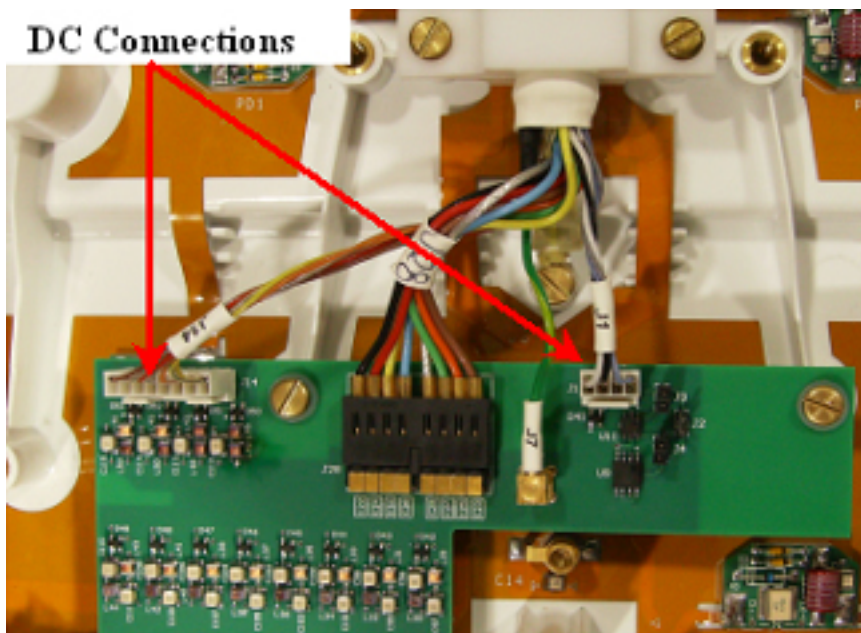
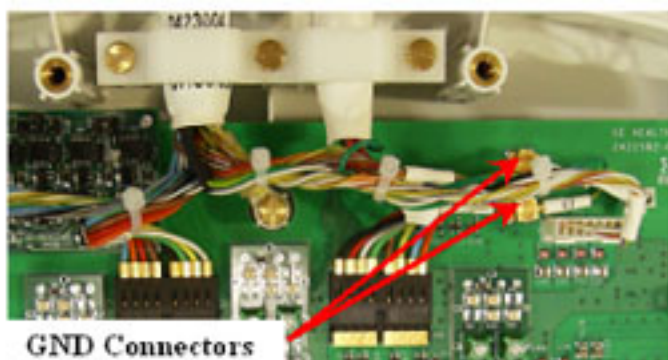


Illustration 5-16: DC Connections: TLU

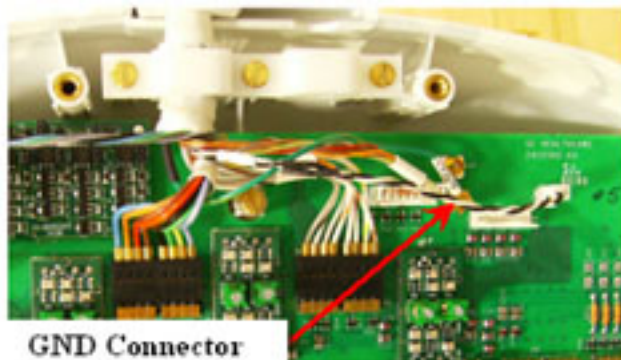


6. Identify and remove GND connectors from interface board. See [Illustration 5-17](#). Pliers may be necessary for removal of the PCX connectors, as they may be difficult to unplug. This will help prevent accidental breakage of the cables. See [Illustration 5-18](#).

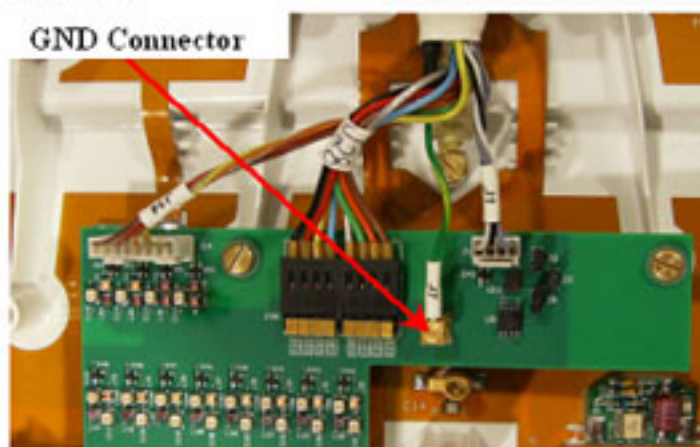
Illustration 5-17: GND Connectors



HNU Upgrade Illustration

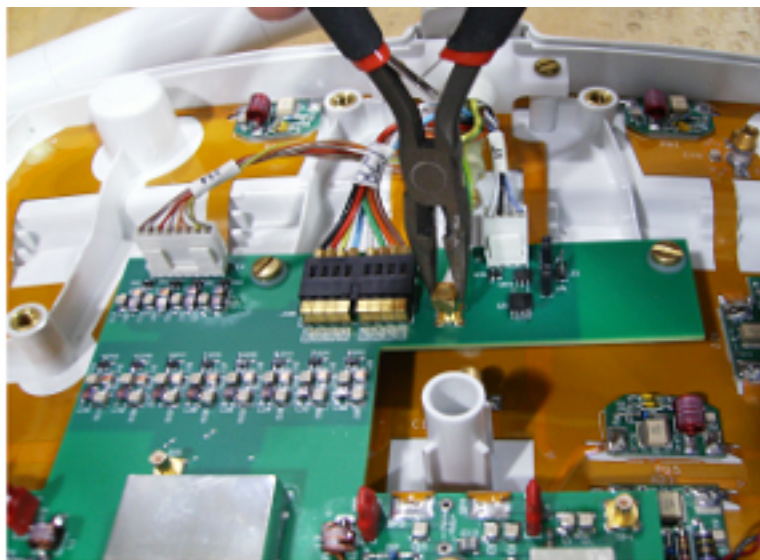


HNU Forward Production & Discovery Illustration



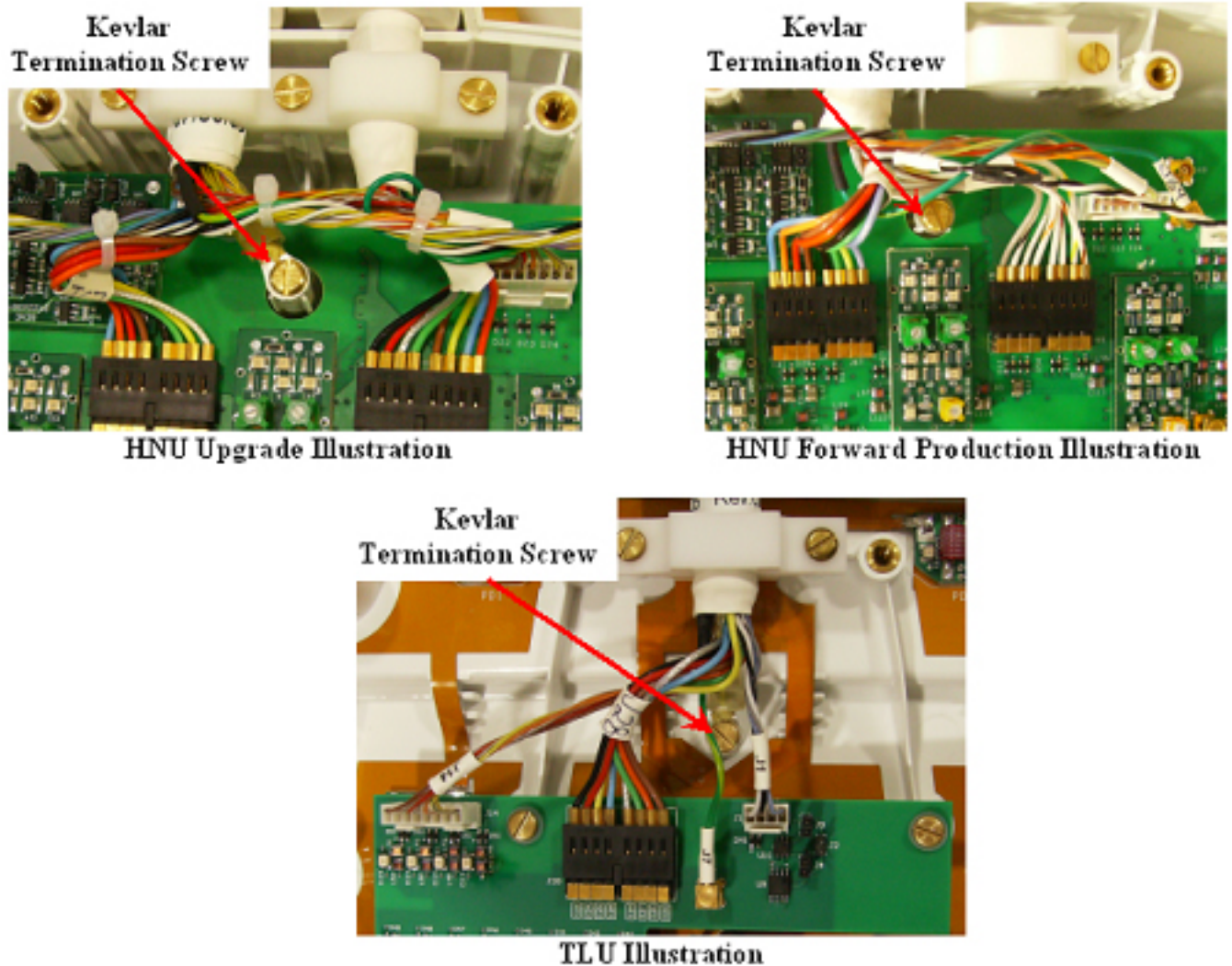
TLU Illustration

Illustration 5-18: Using Pliers For Removal of Connectors



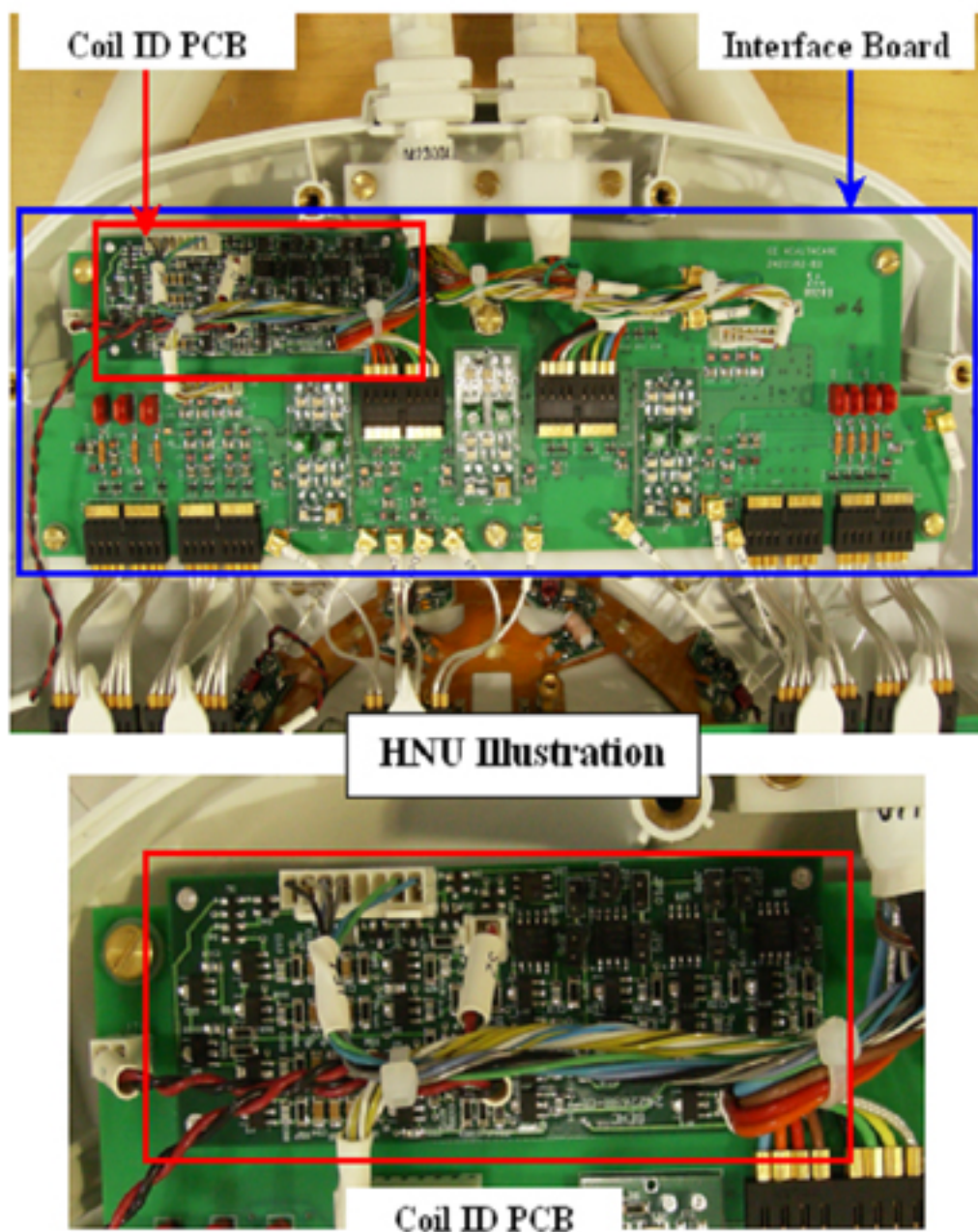
7. Unscrew the Kevlar termination ring as shown in [Illustration 5-19](#). Retain original screw used.

Illustration 5-19: Kevlar Termination Screw



8. Remove the system cable (with or without the coil id).
9. Refer to the Replacement Parts section, if the cable FRU is with the coil id; remove the coil ID PCB from Interface Board (proceed to the next step). If the cable FRU exists without the coil id board, then remove only the cable.
10. Identify the Interface Board Assembly and the coil ID PCB located on the Interface Board (Head Neck Unit (HNU) Only). Carefully remove the Coil ID assembly. See [Illustration 5-20](#).

Illustration 5-20: Interface Board and Coil ID PCB Locations



11. Install the new system cable.

12. If applicable from Step 4: Mount the new coil ID PCB assembly in the open slot on the Interface Board (HNU Only). See [Illustration 5-20](#).

13. Align the cable clamp of the cable into the notch of former slot. Screw down cable clamp using original screws. (Cable clamp screws are 12 mm in length for the TLU and 10 mm in length for the HNU) See [Illustration 5-11](#).
14. Connect RF connectors to Interface Board. Verify that all labels on the connectors match those on the Interface Board. See [Illustration 5-12](#).
15. Connect DC connectors to Interface Boards and coil ID PCB (Coil ID PCB for HNU Only). DC connector "S2" connects J12 located on the 11 channel feedboard to J56 located on the coil ID PCB. DC connector "S1" connects J70 located on the Interface Board to J55 located on the coil ID PCB. Verify that all labels on the connectors match those on the Interface Board. See [Illustration 5-14](#).
16. Connect GND connectors to the Interface Board. Verify that all labels on the connectors match those on the Interface Board. See [Illustration 5-17](#).
17. Screw down Kevlar termination ring using original screw. (Kevlar termination screws are 10mm in length for both the TLU and HNU) See [Illustration 5-19](#).
18. For forward production coils, ensure that jumpers (red suitcase) J5, J40, J41, J55, J28, J8, and J39 are removed from cable interface (MUX) board before scanning.
19. Replace cover. Use original screws to fasten cover.

1.5 Finalization

1. Perform a good bye scan with this coil to confirm that the repaired coil operates properly.

2 1.5T Head Neck Spine (HNS) Mirror Repair/Replacement

2.1 Personnel Requirements

Personnel Requirements	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	40 mins	Not Applicable

2.2 Overview

Follow this process to reassemble the mirror assembly.

2.3 Preliminary Requirements

2.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
1/8" and 1/4" Flathead Screwdrivers	1 each	Replacement of mirror assembly	-	-

2.3.2 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Mirror assembly FRU	1	Replacement of mirror assembly	2419521	GEHC Coils

2.4 Procedure

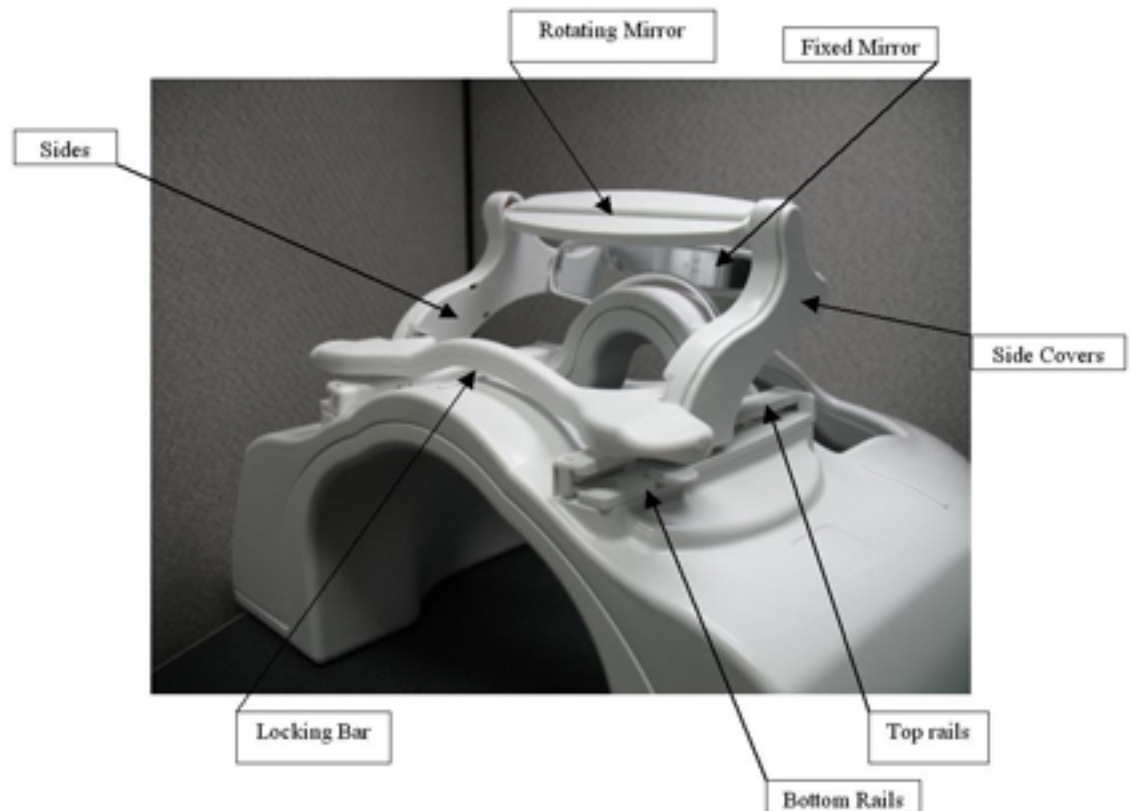


NOTICE

Please refer to the Mirror assembly FRU list above in Replacement Parts.

FRU is needed only if the mirror assembly cannot be properly reassembled.

Illustration 5-21: Parts of the Mirror



1. Inspect mirrors to ensure that they are not coming loose from the plastic housings. If the mirrors are coming loose a new mirror assembly will be needed.
2. Inspect all parts for broken snap features. If any of the snap features are broken, a new mirror assembly will be needed.
 - a. There are 2 located on each end of each of the cross pieces (the locking bar, the rotating mirror, and the fixed mirror). See [Illustration 5-22](#).

Illustration 5-22: Cross-Piece Snap Features



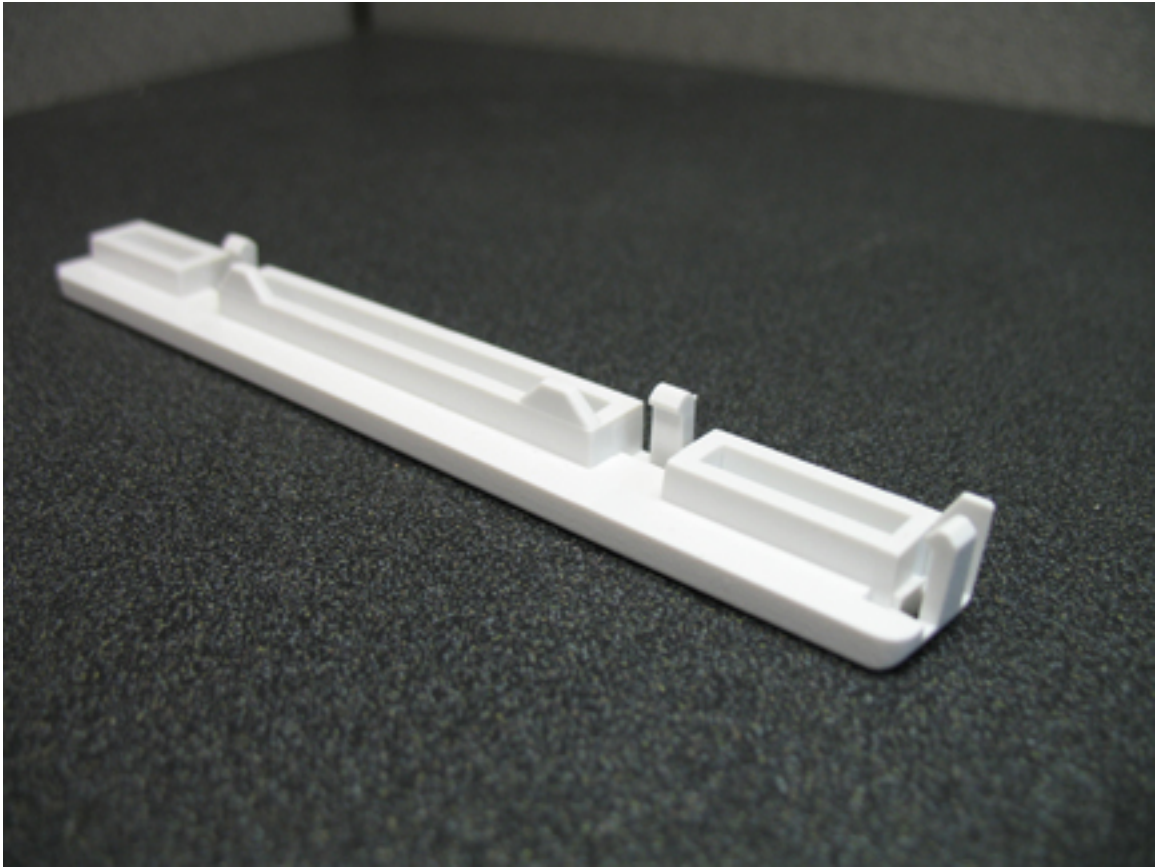
- b. There are 5 on each of the side covers. See [Illustration 5-23](#).

Illustration 5-23: Snap Features on Side Covers



- c. There are 3 on each of the top rails. See [Illustration 5-24](#).

Illustration 5-24: Snap Features on Top Rails



3. If any of the cross pieces have been dislocated from their proper position (see [Illustration 5-25](#)), the side covers will need to be removed to properly re-seat the snap feature. This is due to the locking features on the inside of the side covers. The locking features are the three round features shown in [Illustration 5-26](#).

Illustration 5-25: Cross-Piece Dislocated



Illustration 5-26: Locking Features on the Side Covers



4. If needed, the side covers can be removed by inserting a small screwdriver from the backside of the sides to disengage the snap feature on the side covers. (See [Illustration 5-27](#).) Start at the bottom and work up to the top of the mirror.

Illustration 5-27: Removing Side Covers



5. When reassembling the cross pieces with the side it is best to engage all three cross pieces at the same time, rather than one at a time.
6. The rotating mirror requires some additional pressure to properly engage. This is because of the o-rings, which supply the correct amount of drag to resist rotation. In the picture shown (see [Illustration 5-28](#)), firm pressure is being applied using the thumb and middle finger while gentle pressure is being applied with the index finger.

Illustration 5-28: Engaging the Rotating Mirror

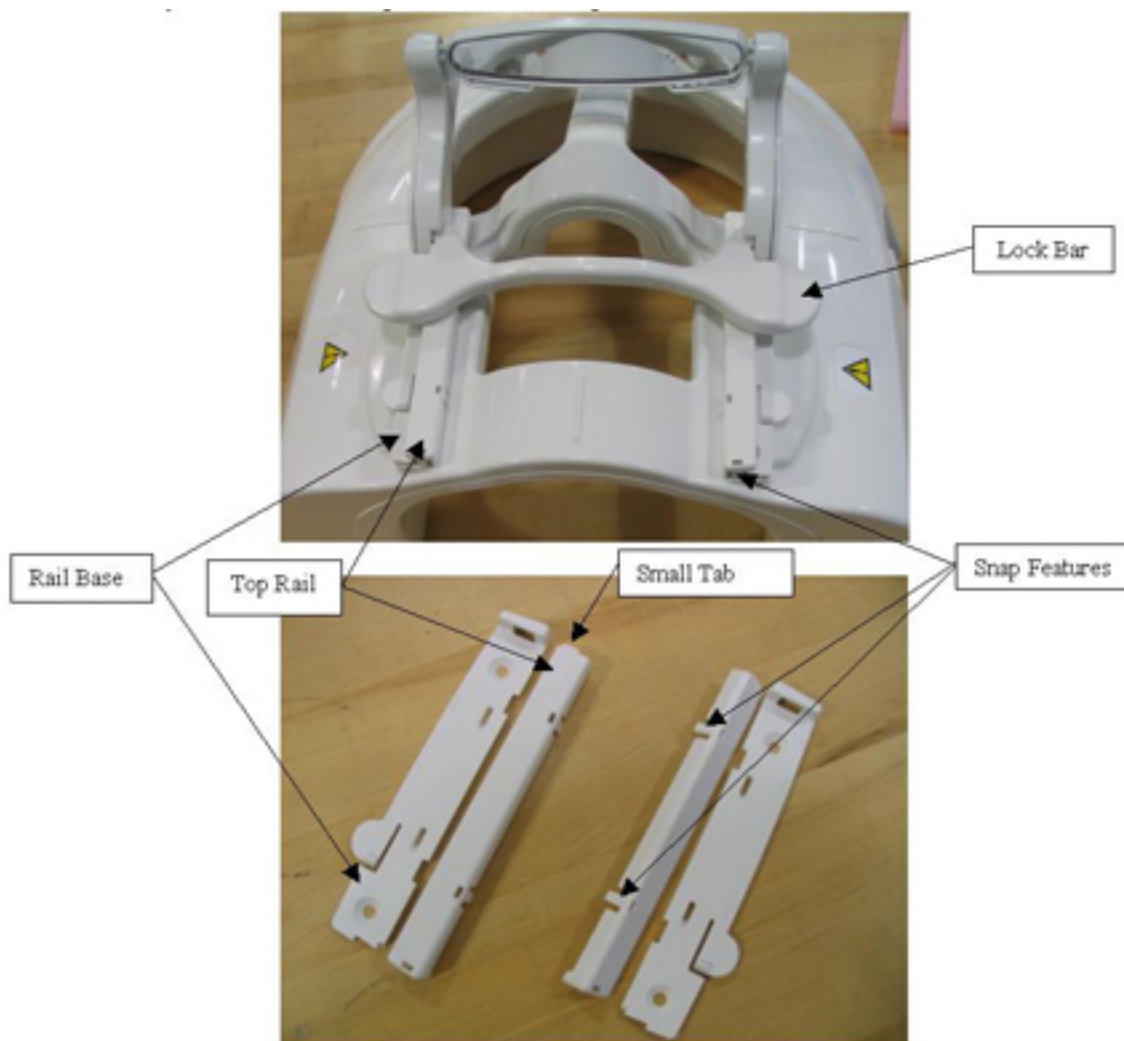


7. The reassembly of the rails below is from the operators manual. To prevent breakage, the mirror is designed to break away from the anterior face component. In the event that the mirror assembly is pulled from the anterior face component, the mirror can be reinstalled using the following procedure:
 - a. Release the lock bar on the mirror assembly, so that the two top rail sections come loose from the mirror assembly.
 - b. Insert the small tab on the top rail in the mirror rail base and snap the rail down so that the three snap features engage the rail base. Repeat for the other side.

NOTE: There are right and left sides which are different. See [Illustration 5-29](#).

 - c. Slide the mirror assembly back onto the rails.

Illustration 5-29: Installing Mirror Rails



8. If a new mirror assembly needs to be installed the FRU kit includes the top and bottom rails as well as the screws needed. A decision can be made about whether the rails need to be replaced, depending on the condition of the old rails. If they are to be replaced the top rails should be pulled off, and the four screws should be removed. Use the new screws to attach the new lower rails torque to 3 inch-pounds. Install top rails as described in [Step 7](#).

2.5 Finalization

Ensure that all joints are firmly seated and that the mirror assembly is functioning properly.

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